Terra Ceia Aquatic Preserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

Contents

Funding & Acknowledgements

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Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

Parameter Name	*Units*	*Low Threshold*	*High Threshold*
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

Parameter Name	*Units*	*Low Threshold*	*High Threshold*
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

Parameter Name	*Units*	*Low Threshold*	*High Threshold*
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m^-1	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

SEACAR QAQC Description	*Include*	*SEACAR QAQCFlagCode*
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

Qualifier Source	*Value Qualifier*	*Include*	*MDL*	*Description*
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determiniation; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

Qualifier Source	*Value Qualifier*	*Include*	*Description*
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as El Niño/La Niña, large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p >= 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Uncorrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

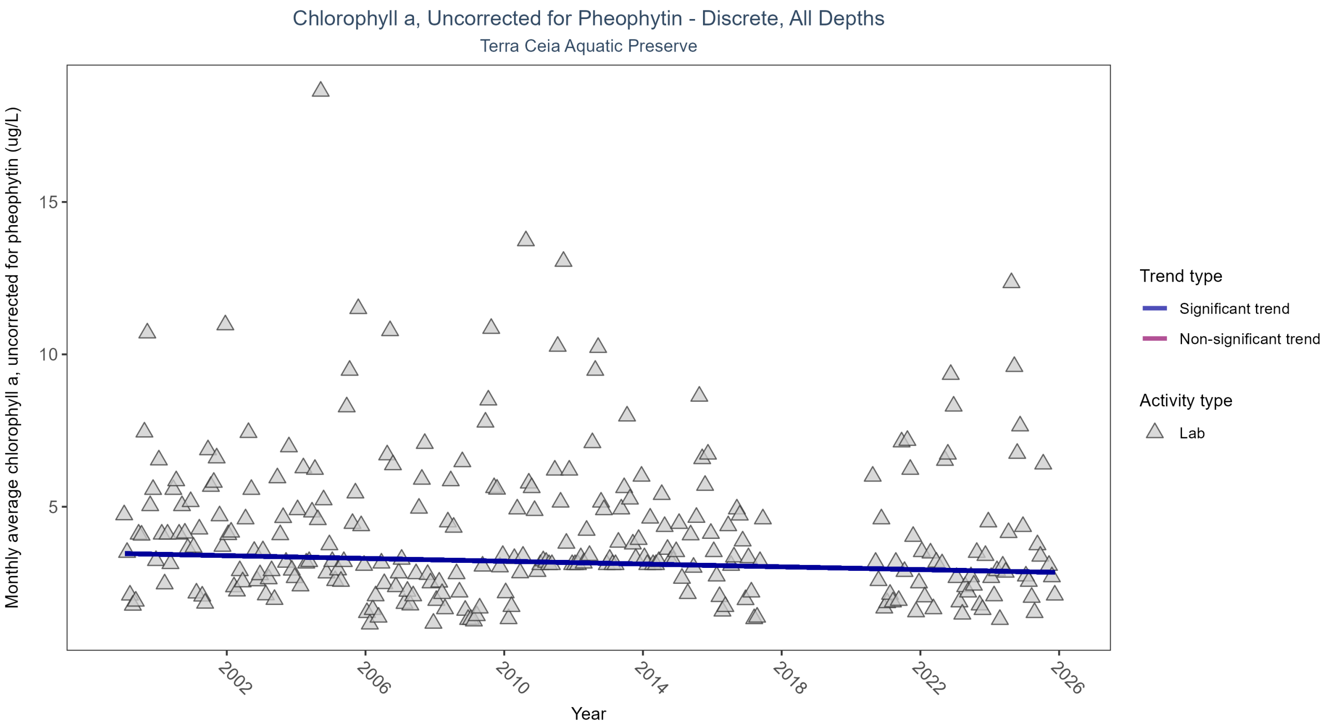

Figure 1: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	1093	25	1999 - 2025	3.2	-0.0933	3.4639	-0.0229	0.029

Monthly average chlorophyll a, uncorrected for pheophytin, decreased by 0.02 µg/L per year, indicating an increase in water clarity.

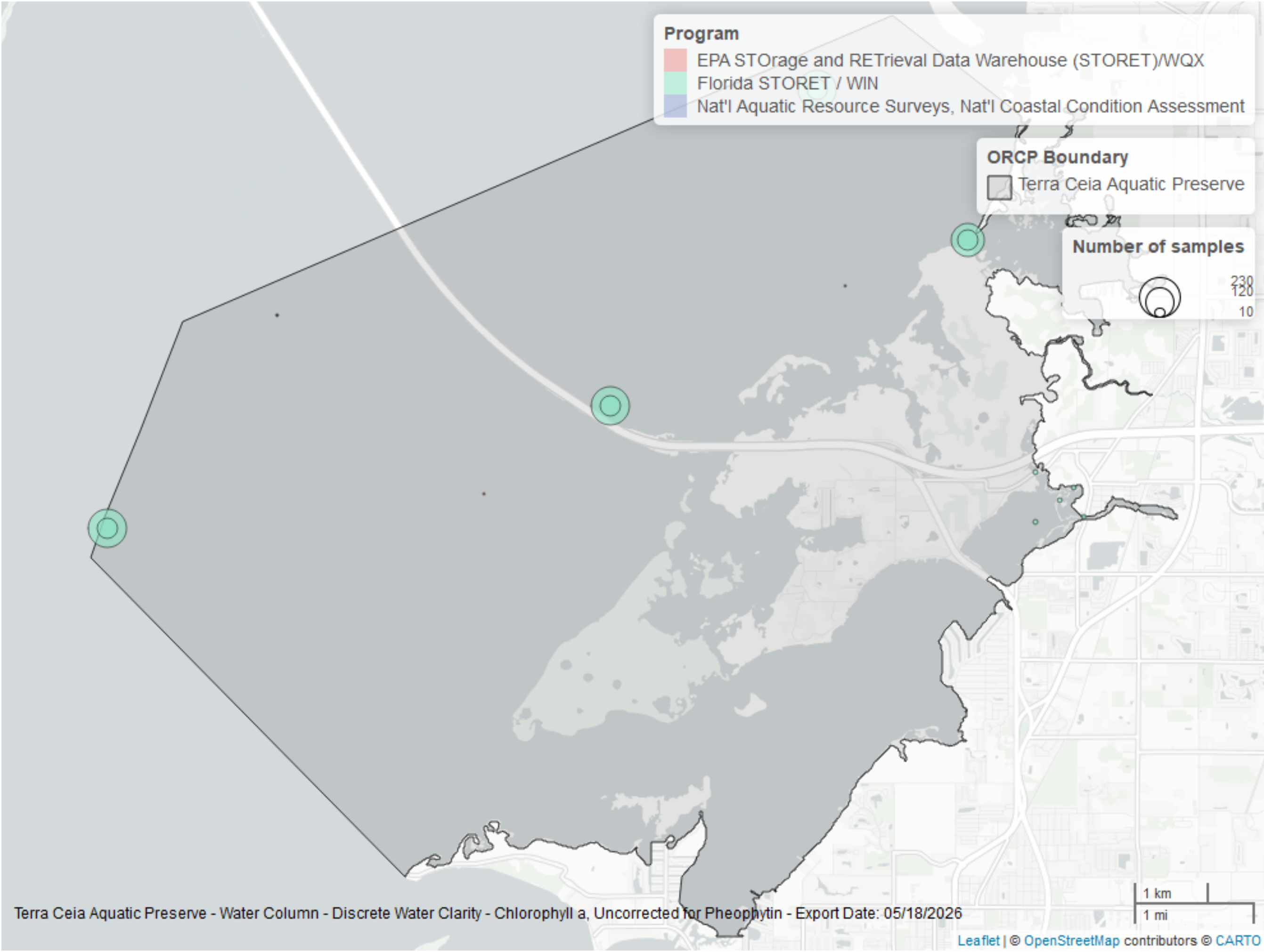

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Terra Ceia Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
5002	1097	1999	2025
103	3	2000	2015
118	1	2010	2010

Program names:

5002 - Florida STORET / WIN[1]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[2]
118 - National Aquatic Resource Surveys, National Coastal Condition Assessment[3]

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

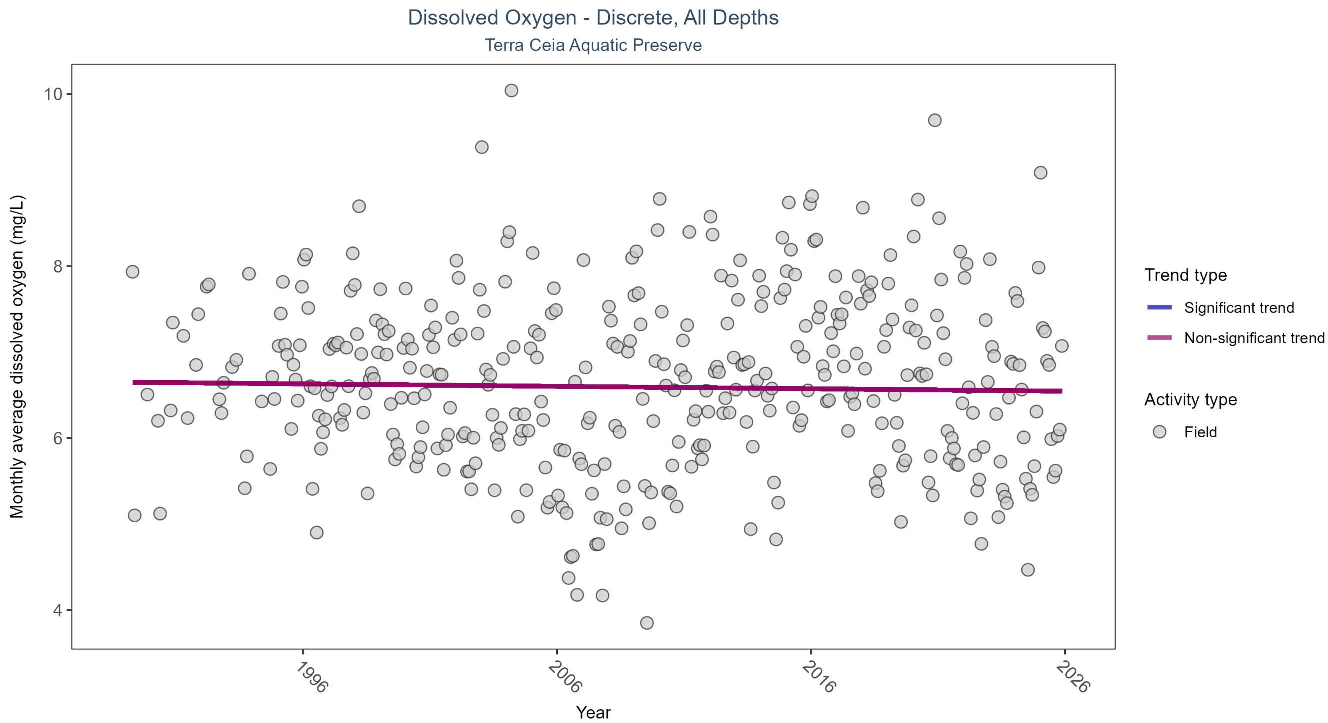

Figure 3: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	25110	37	1989 - 2025	6.52	-0.0277	6.649	-0.0028	0.4223

Dissolved oxygen showed no detectable trend between 1989 and 2025.

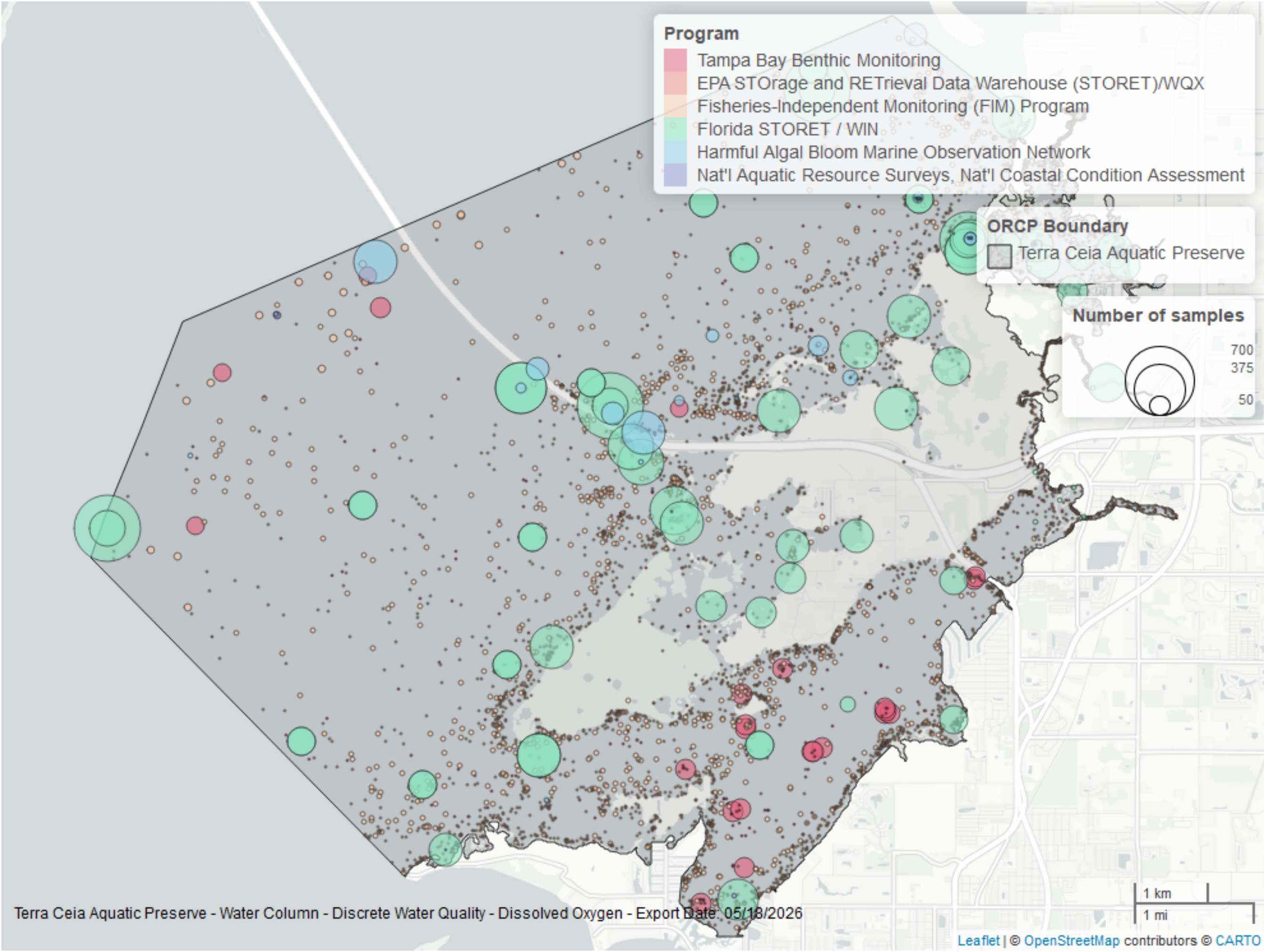

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Terra Ceia Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
5002	11590	1995	2025
69	10597	1989	2024
4067	1869	1993	2023
95	1260	1999	2018
118	19	2015	2020
103	3	2015	2015

Program names:

5002 - Florida STORET / WIN[1]
69 - Fisheries-Independent Monitoring (FIM) Program[4]
4067 - Tampa Bay Benthic Monitoring[5]
95 - Harmful Algal Bloom Marine Observation Network[6]
118 - National Aquatic Resource Surveys, National Coastal Condition Assessment[3]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[2]

Dissolved Oxygen Saturation - Discrete

Seasonal Kendall-Tau Trend Analysis

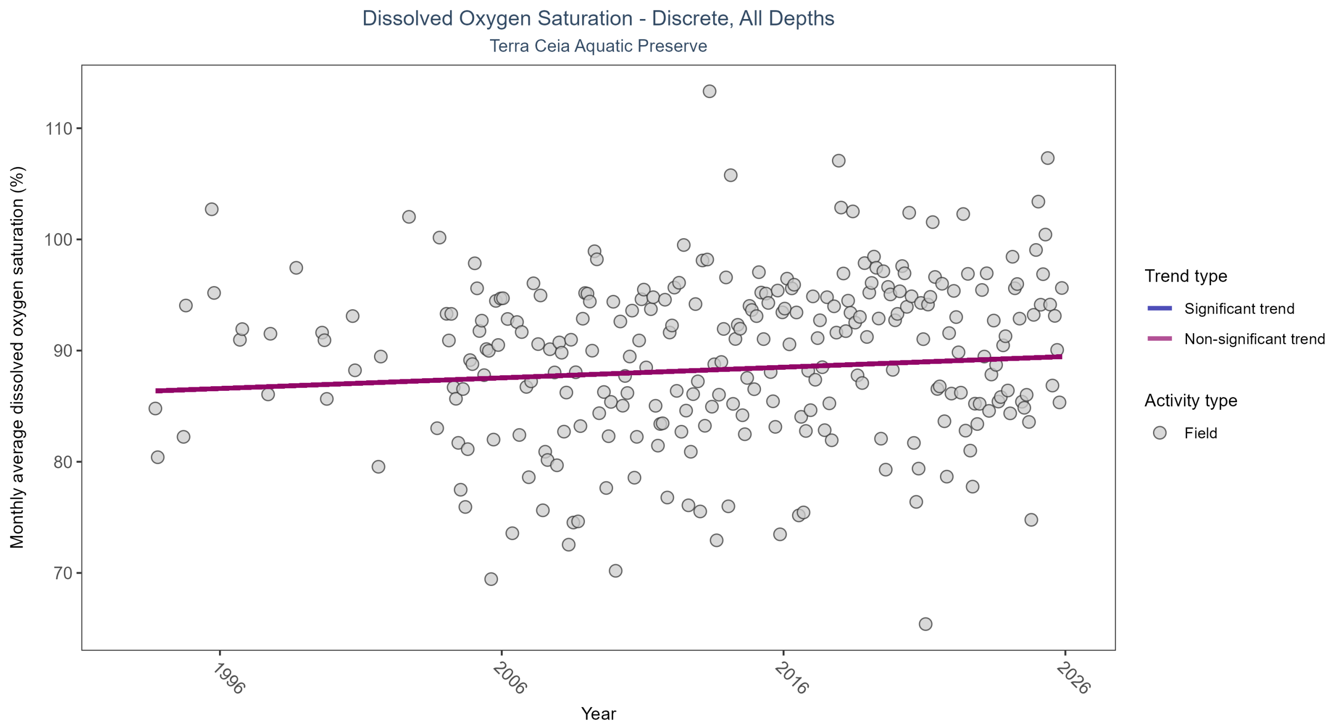

Figure 5: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	5224	33	1993 - 2025	90.3	0.0955	86.3011	0.0958	0.1081

Dissolved oxygen saturation showed no detectable trend between 1993 and 2025.

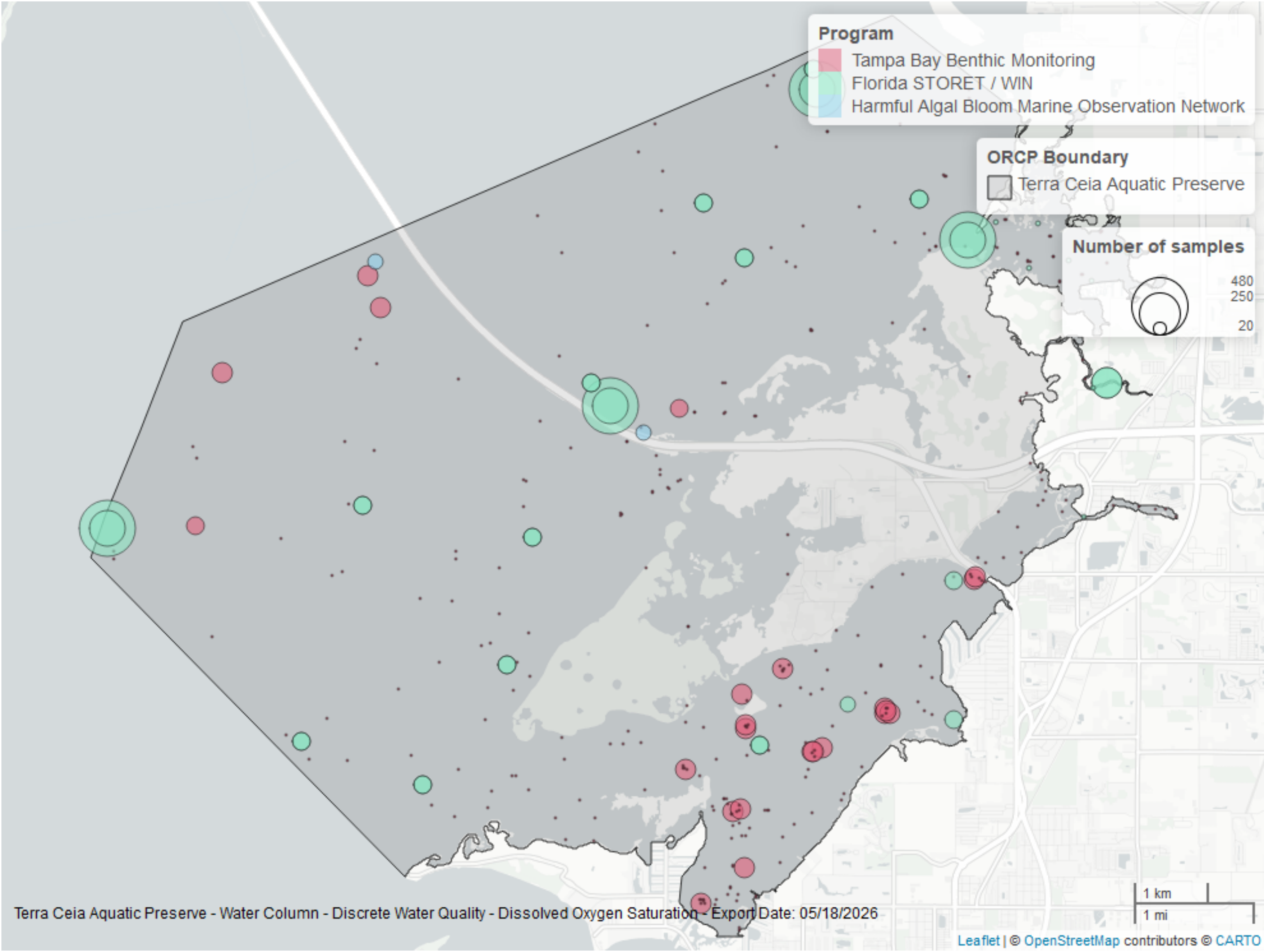

Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Terra Ceia Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Dissolved Oxygen Saturation

ProgramID	N_Data	YearMin	YearMax
5002	3491	2004	2025
4067	1894	1993	2023
95	67	2014	2018

Program names:

5002 - Florida STORET / WIN[1]
4067 - Tampa Bay Benthic Monitoring[5]
95 - Harmful Algal Bloom Marine Observation Network[6]

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

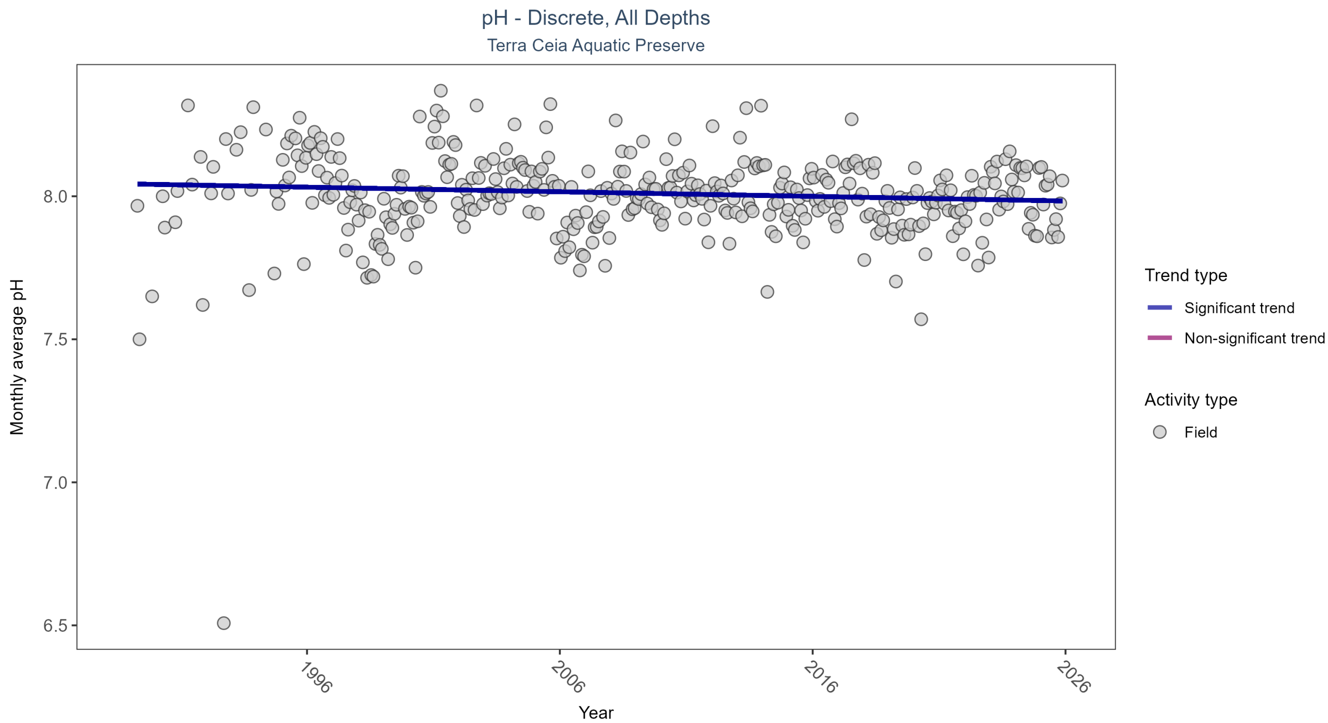

Figure 7: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	23347	37	1989 - 2025	8	-0.0949	8.0431	-0.0016	0.0085

Monthly average pH decreased by less than 0.01 pH units per year.

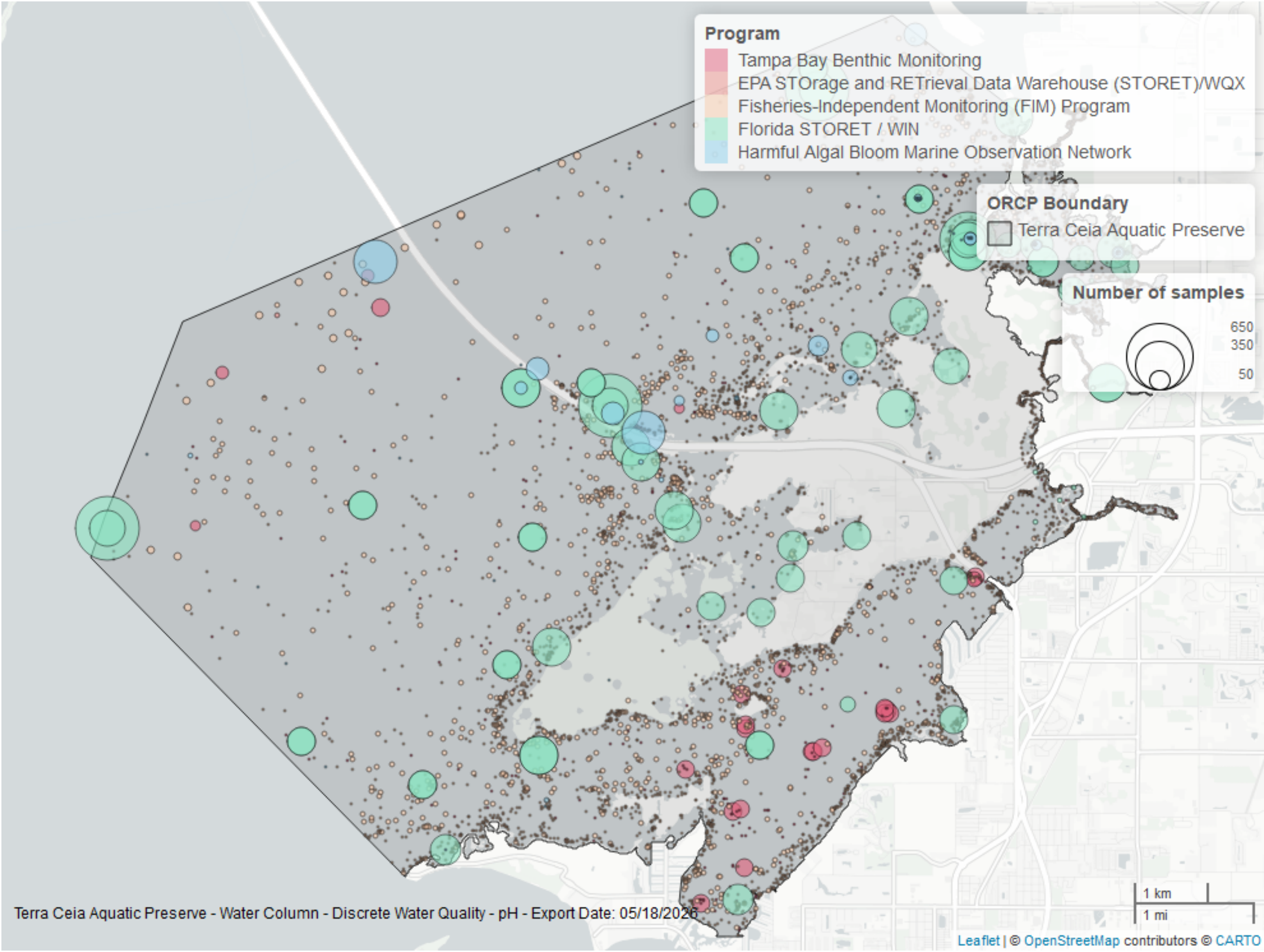

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Terra Ceia Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
69	10503	1989	2024
5002	10060	1995	2025
4067	1596	1993	2023
95	1255	1999	2018
103	5	2015	2015

Program names:

69 - Fisheries-Independent Monitoring (FIM) Program[4]
5002 - Florida STORET / WIN[1]
4067 - Tampa Bay Benthic Monitoring[5]
95 - Harmful Algal Bloom Marine Observation Network[6]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[2]

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

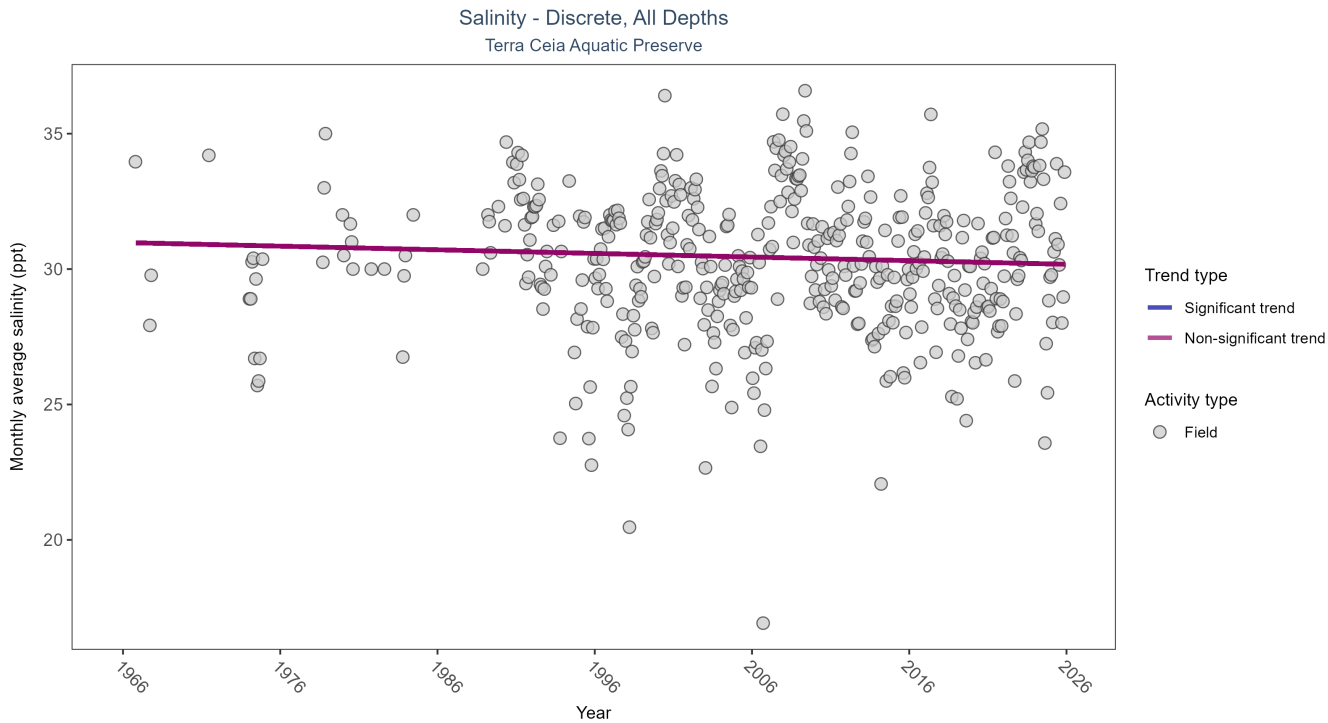

Figure 9: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No detectable trend	25494	49	1966 - 2025	30.6	-0.0482	30.9801	-0.0134	0.144

Salinity showed no detectable trend between 1966 and 2025.

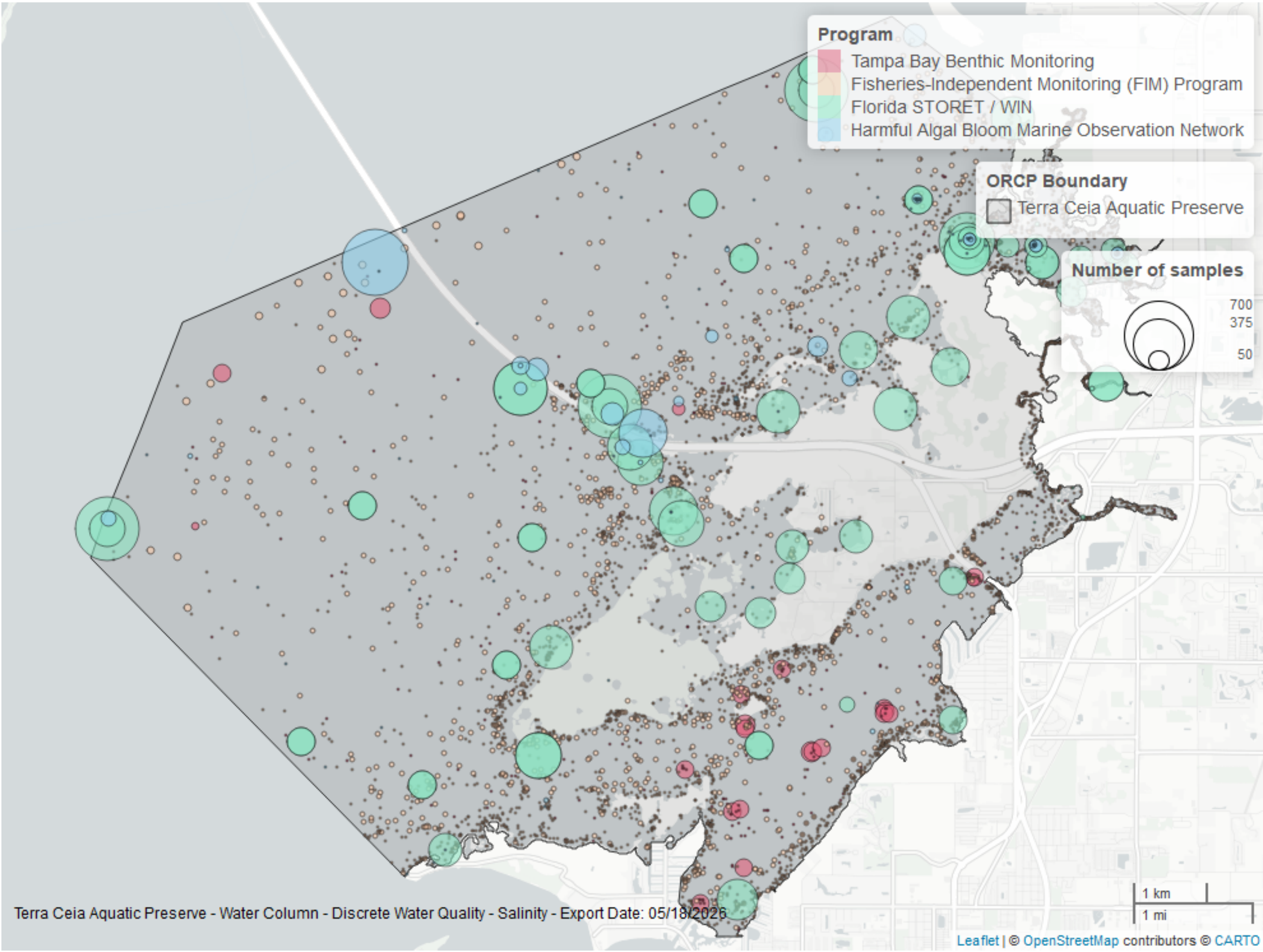

Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Terra Ceia Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
5002	11282	1995	2025
69	10660	1989	2024
95	1976	1966	2018
4067	1609	1993	2023

Program names:

5002 - Florida STORET / WIN[1]
69 - Fisheries-Independent Monitoring (FIM) Program[4]
95 - Harmful Algal Bloom Marine Observation Network[6]
4067 - Tampa Bay Benthic Monitoring[5]

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

1) TN = TKN + NO3O2;
2) TN = TKN + NO3 + NO2;
3) TN = ORGN + NH4 + NO3O2;
4) TN = ORGN + NH4 + NO2 + NO3;
5) TN = TKN + NO3;
6) TN = ORGN + NH4 + NO3;

Additional Information:

- Rules for use of sample fraction:
  - Florida Department of Environmental Protection (FDEP) report that if both "Total" and "Dissolved" components are reported, only "Total" is used. If the total is not reported, then the dissolved components are used as a best available replacement.
  - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
  - ParameterName = "Total Nitrogen"
  - SEACAR_QAQCFlagCode = "1Q"
  - SEACAR_QAQC_Description = "SEACAR Calculated"

Seasonal Kendall-Tau Trend Analysis

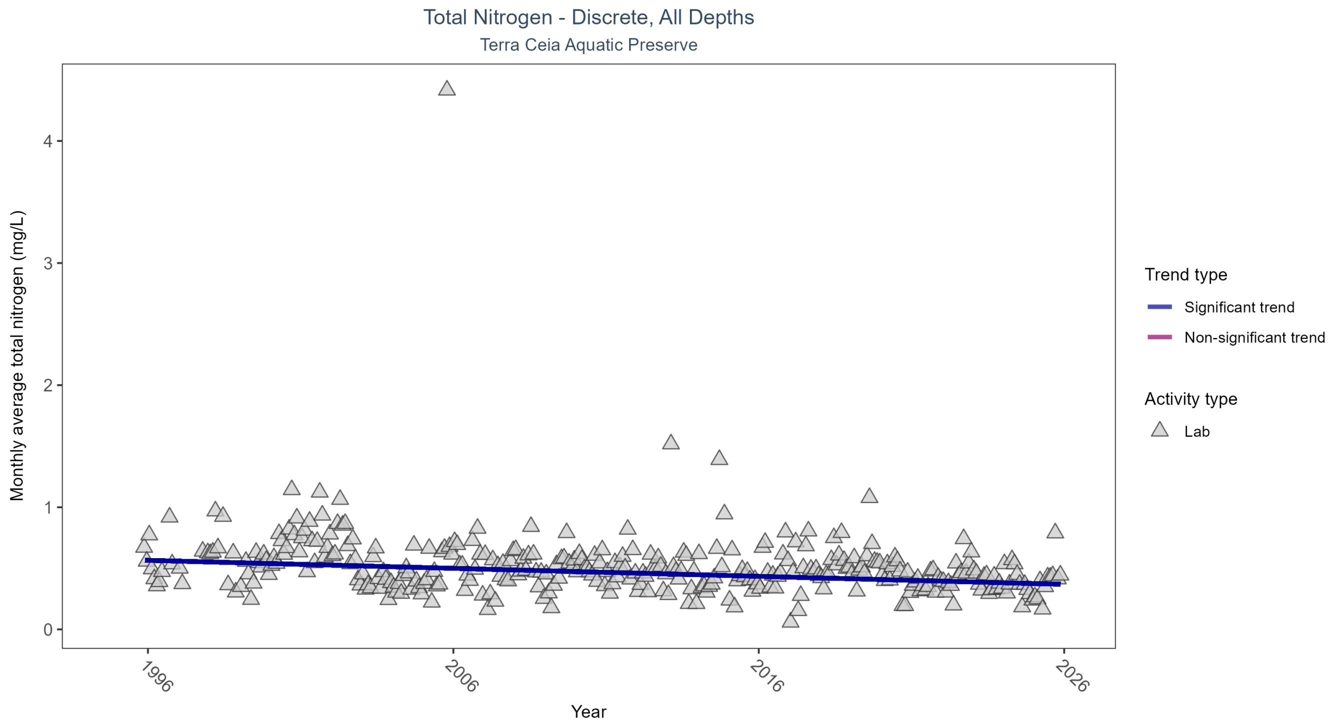

Figure 11: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	2444	31	1995 - 2025	0.3765	-0.243	0.5719	-0.0066	0

Monthly average total nitrogen decreased by 0.01 mg/L per year.

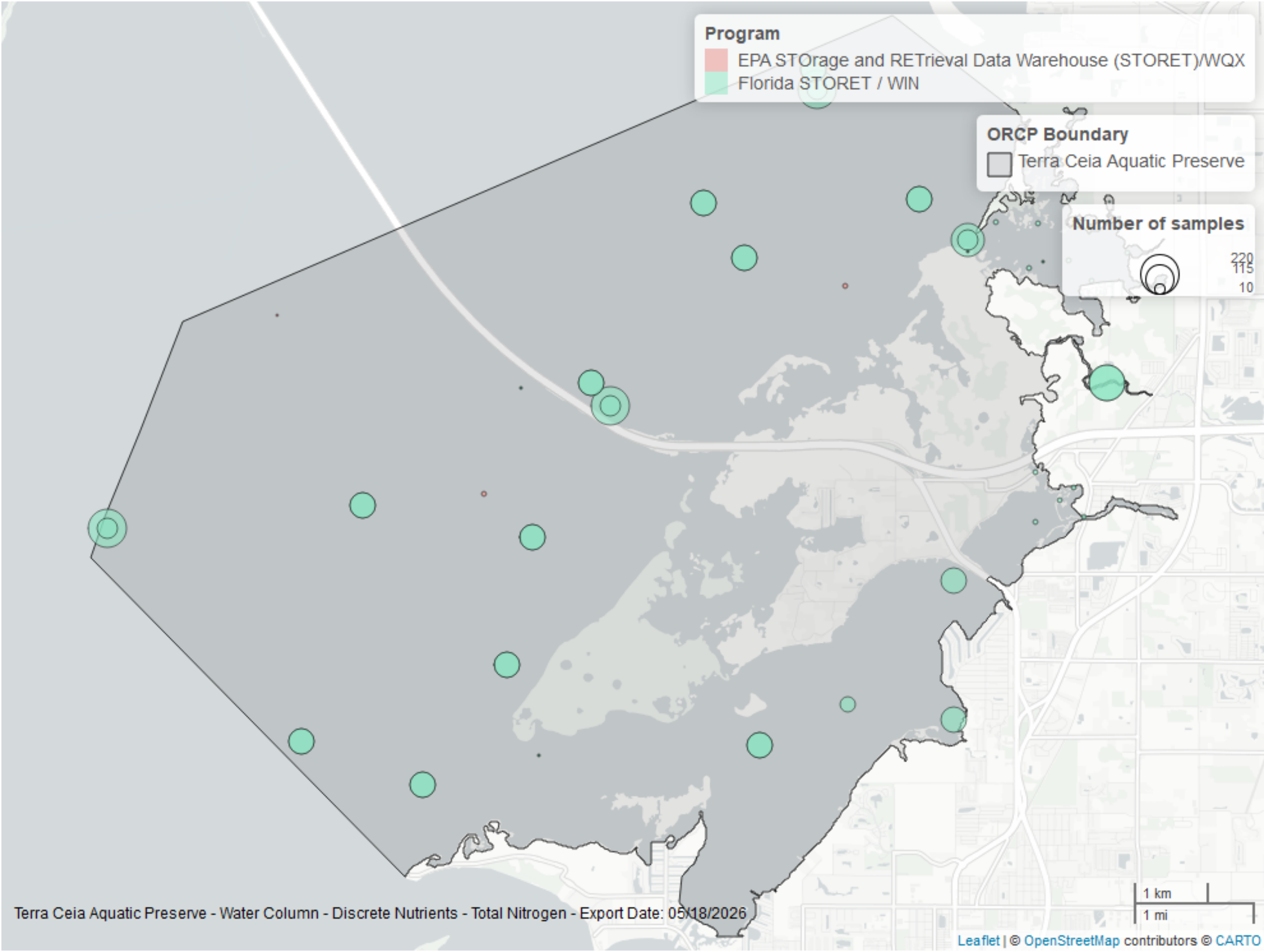

Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Terra Ceia Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
5002	2461	1995	2025
103	7	2000	2015

Program names:

5002 - Florida STORET / WIN[1]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[2]

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

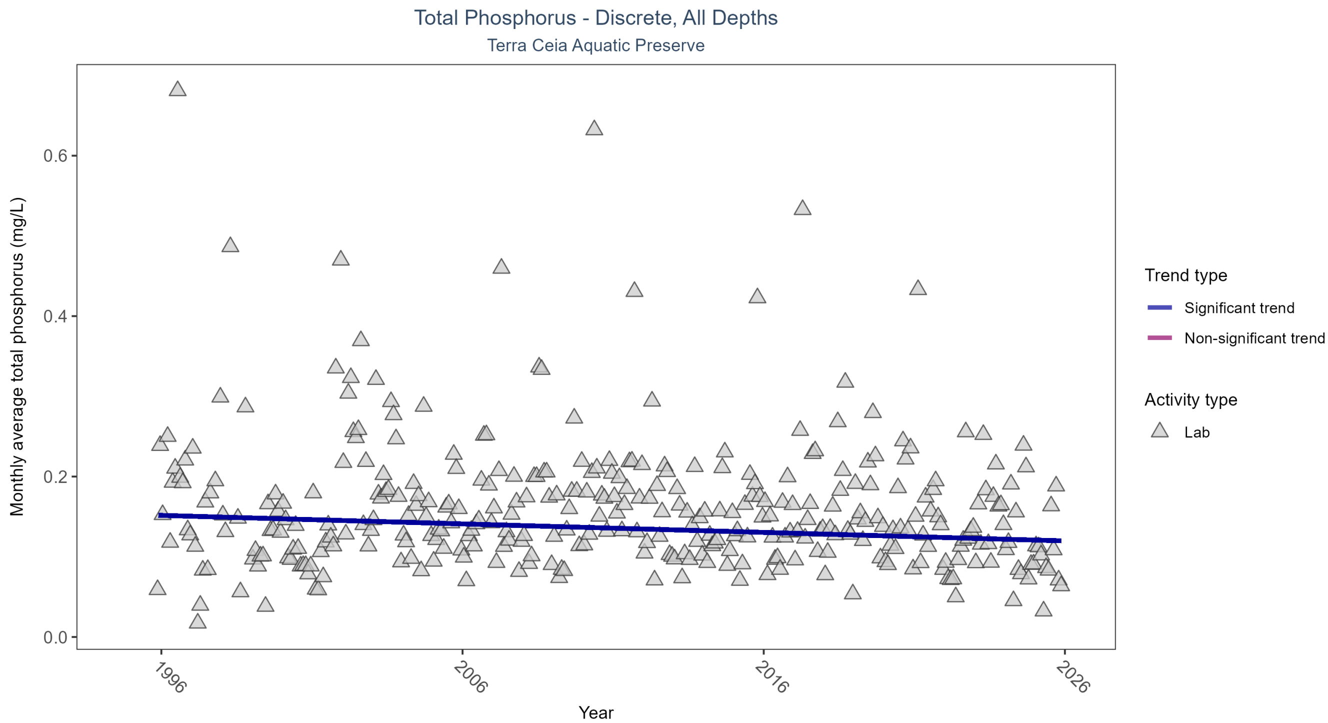

Figure 13: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	3277	31	1995 - 2025	0.106	-0.1081	0.1526	-0.0011	0.0037

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

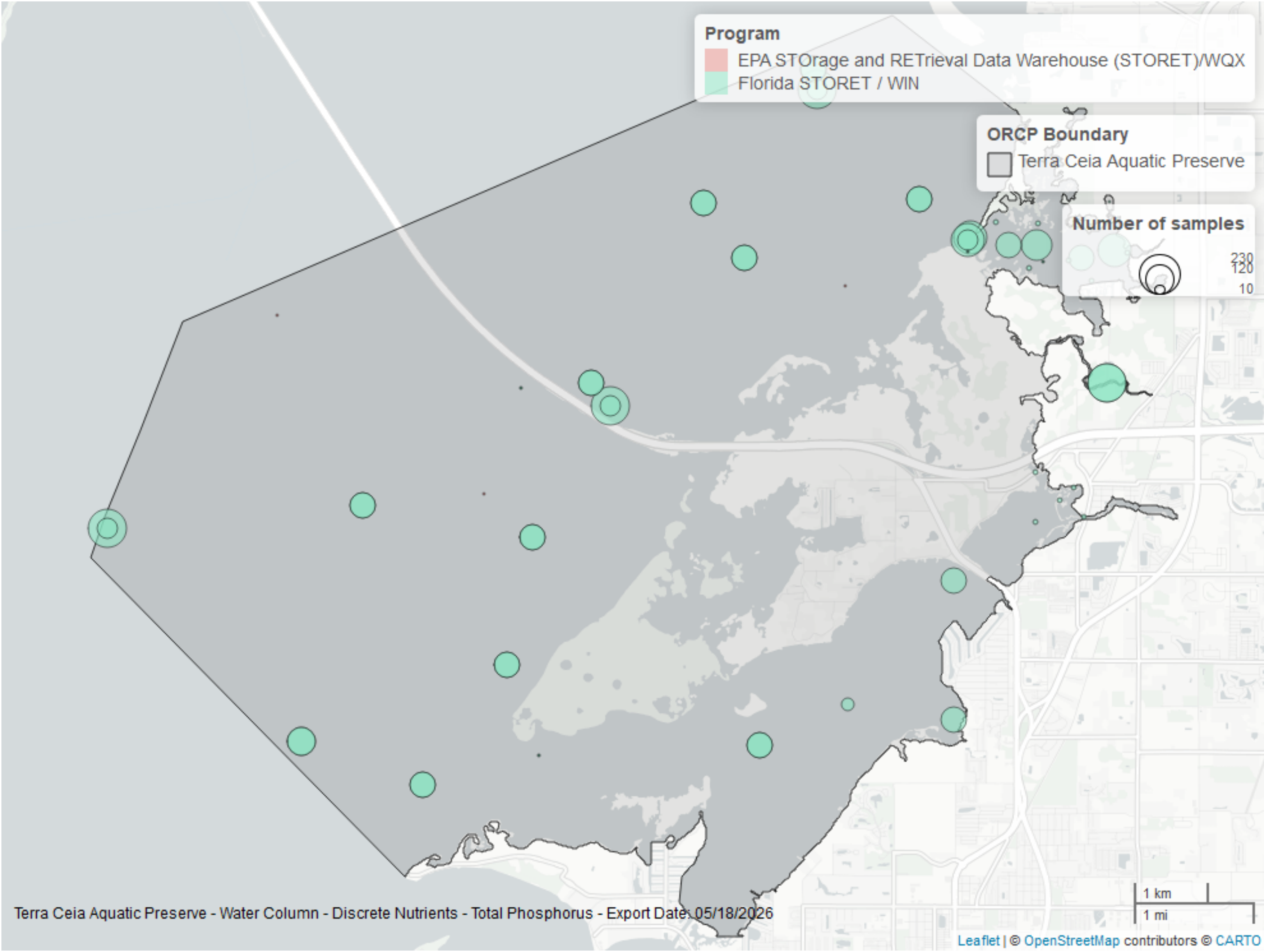

Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Terra Ceia Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
5002	3390	1995	2025
103	5	2000	2015

Program names:

5002 - Florida STORET / WIN[1]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[2]

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

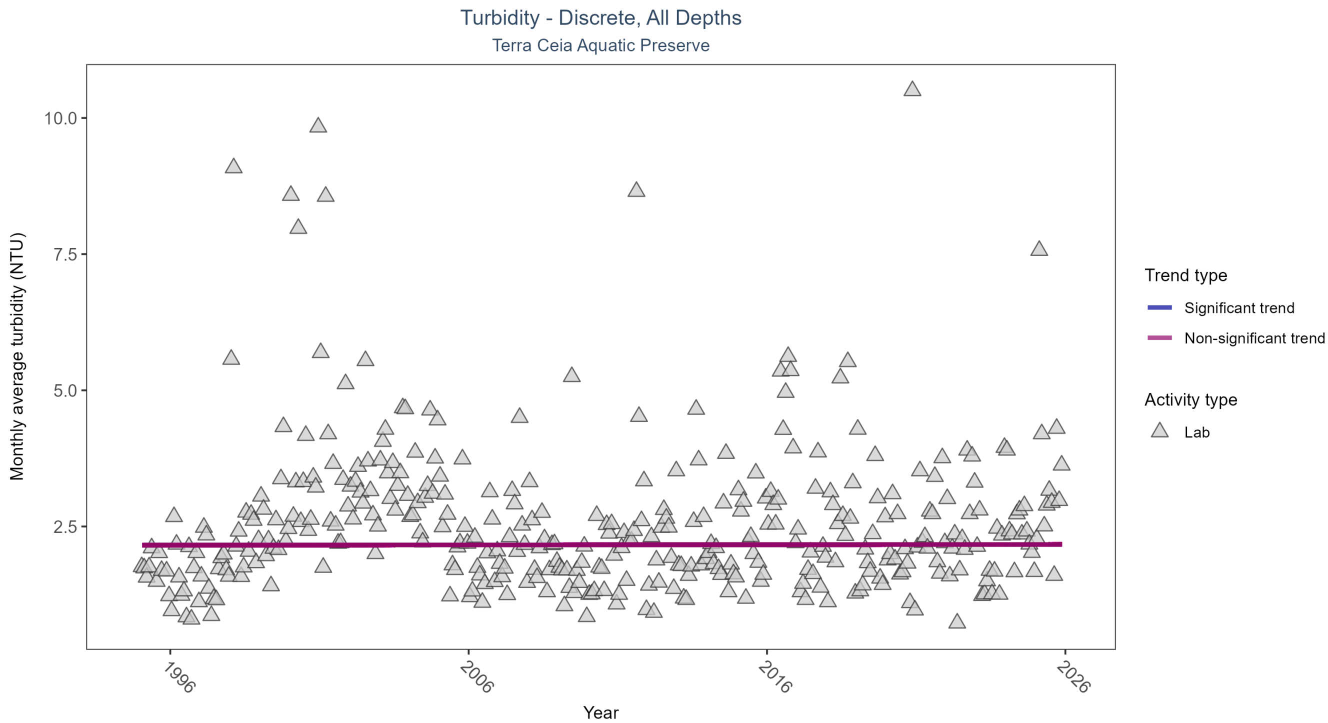

Figure 15: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	8131	31	1995 - 2025	1.9	0.0028	2.1557	0.0005	0.9288

Turbidity showed no detectable trend between 1995 and 2025.

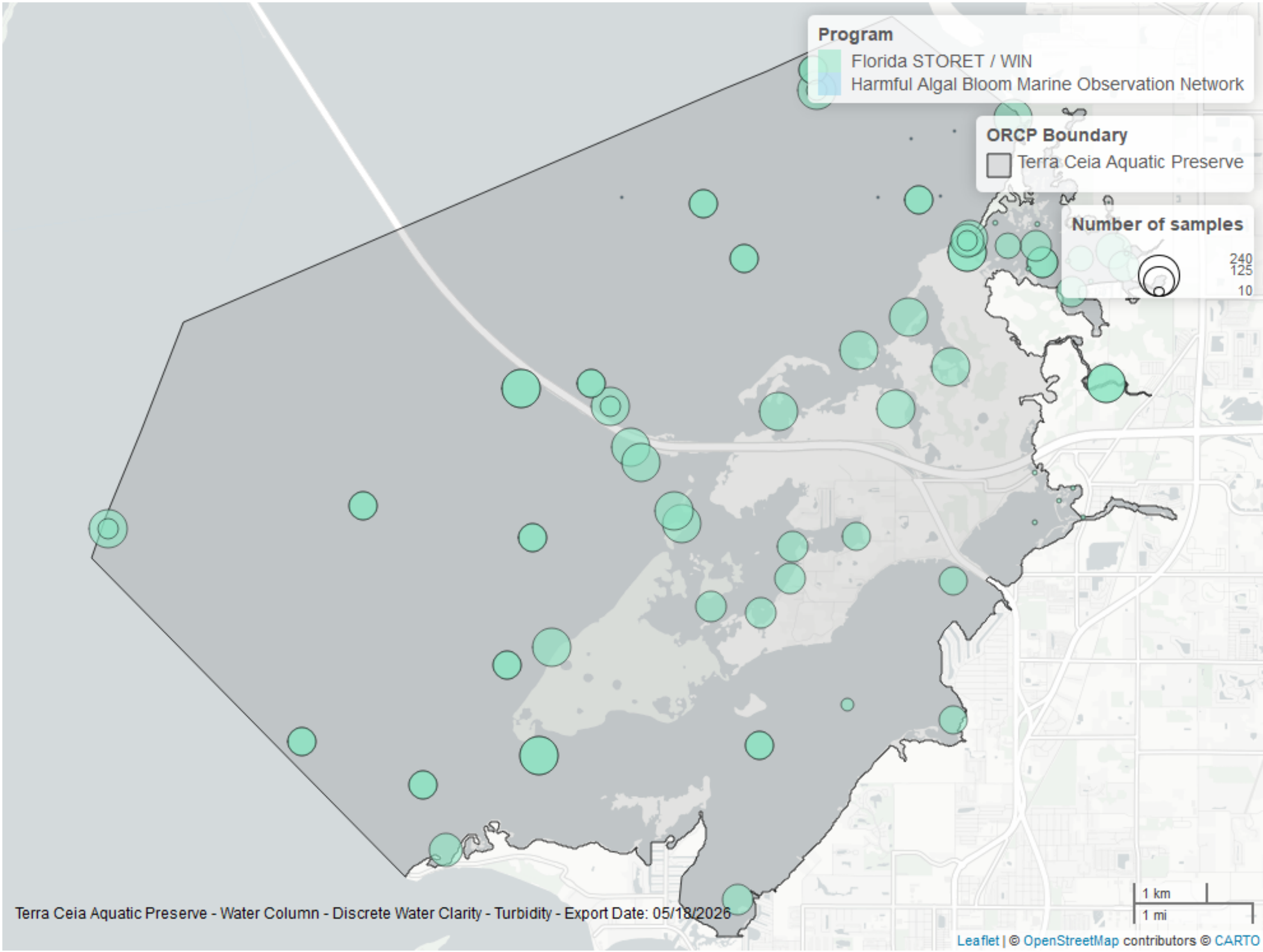

Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Terra Ceia Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
5002	8196	1995	2025
95	8	2002	2004

Program names:

5002 - Florida STORET / WIN[1]
95 - Harmful Algal Bloom Marine Observation Network[6]

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

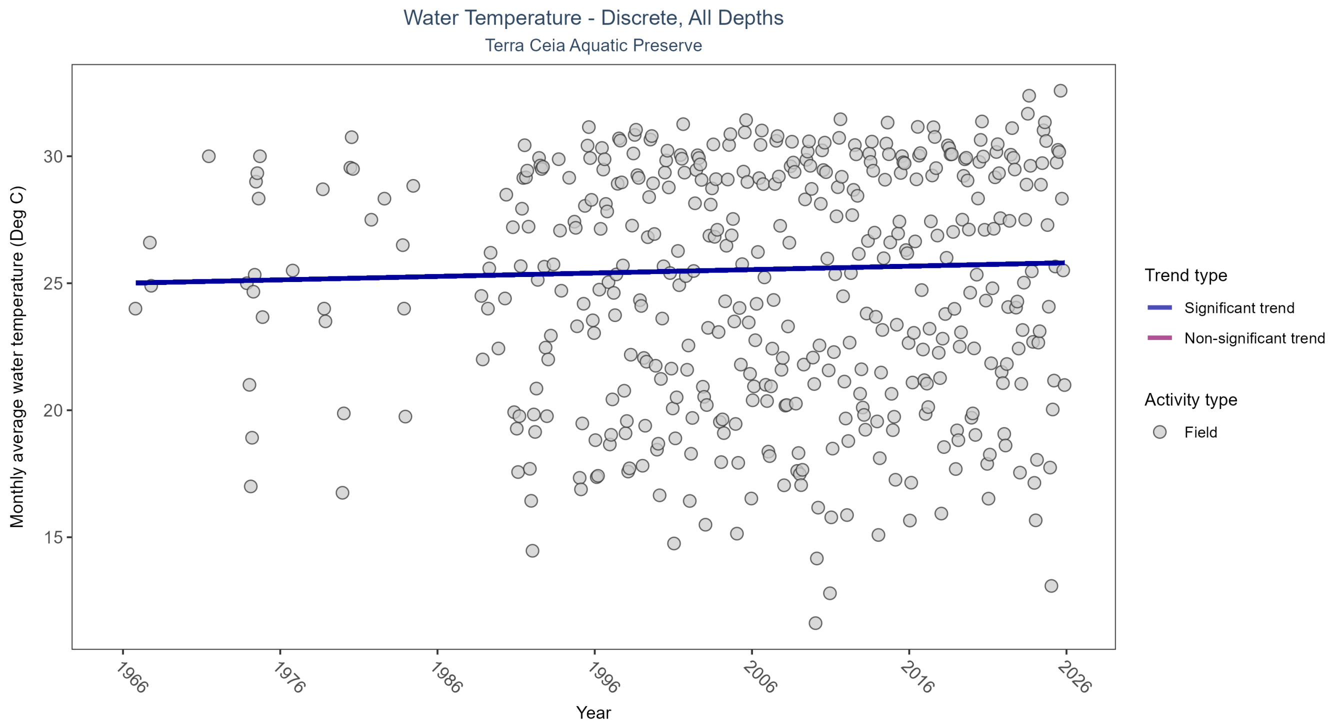

Figure 17: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	26272	51	1966 - 2025	26.7	0.0907	24.9994	0.0134	0.0077

Monthly average water temperature increased by 0.01°C per year.

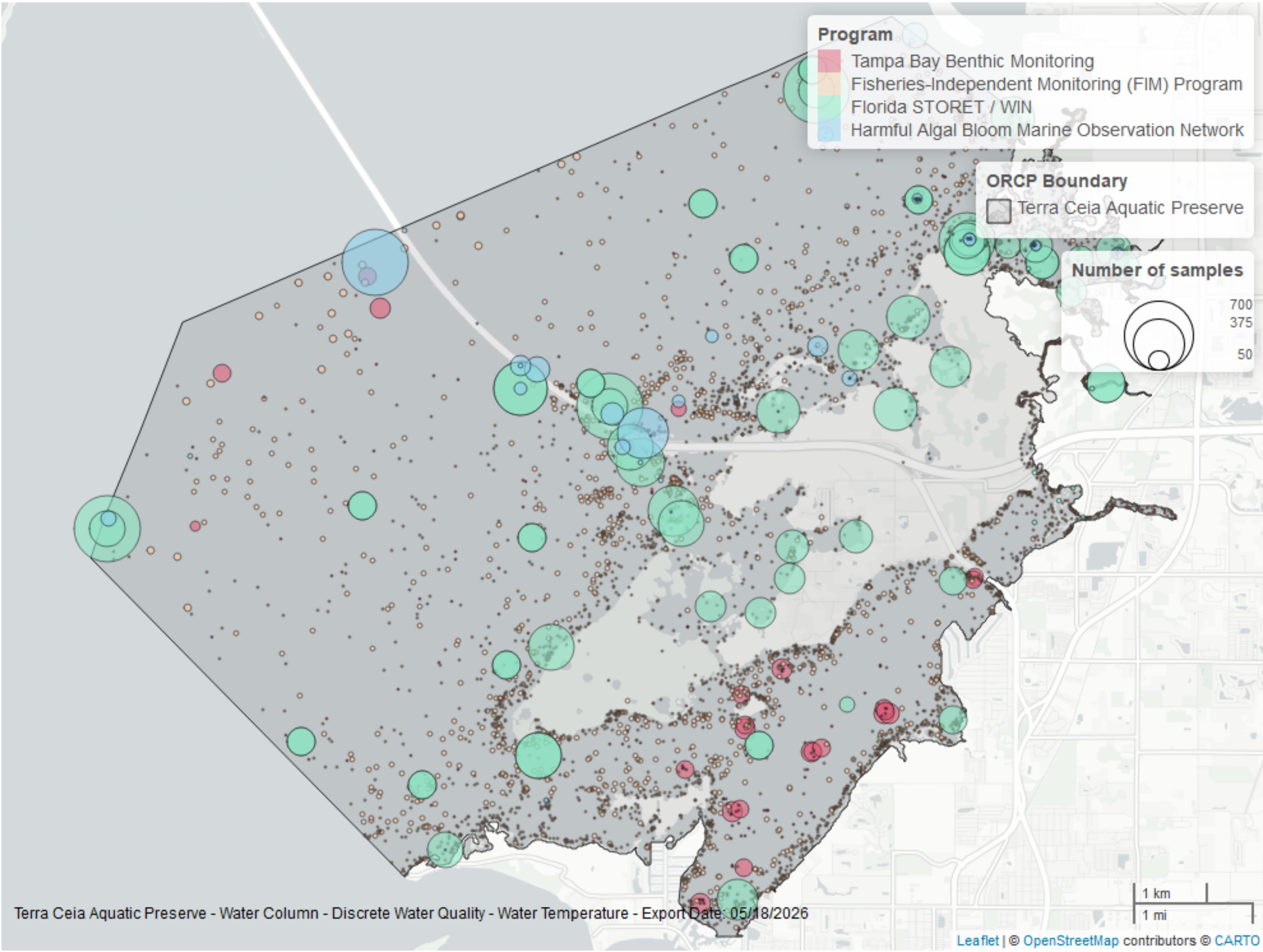

Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Terra Ceia Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
5002	11883	1995	2025
69	10669	1989	2024
95	1971	1966	2018
4067	1775	1993	2023

Program names:

5002 - Florida STORET / WIN[1]
69 - Fisheries-Independent Monitoring (FIM) Program[4]
95 - Harmful Algal Bloom Marine Observation Network[6]
4067 - Tampa Bay Benthic Monitoring[5]

Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

Continuous monitoring locations in Terra Ceia Aquatic Preserve

Table 24: Station overview for Continuous parameters by Program

ProgramID	*ProgramLocationID*	*Years of Data*	*Use in Analysis*	*Parameters*
7	023000825	2	FALSE	Water Temperature , Salinity , Specific Conductivity
7	02300084	2	FALSE	Water Temperature , Salinity , Specific Conductivity
7	023000842	2	FALSE	Water Temperature , Salinity , Specific Conductivity
473	TCBH	2	FALSE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity

Program names:

7 - National Water Information System[7]
473 - Terra Ceia Aquatic Preserve Continuous Water Quality Monitoring[8]

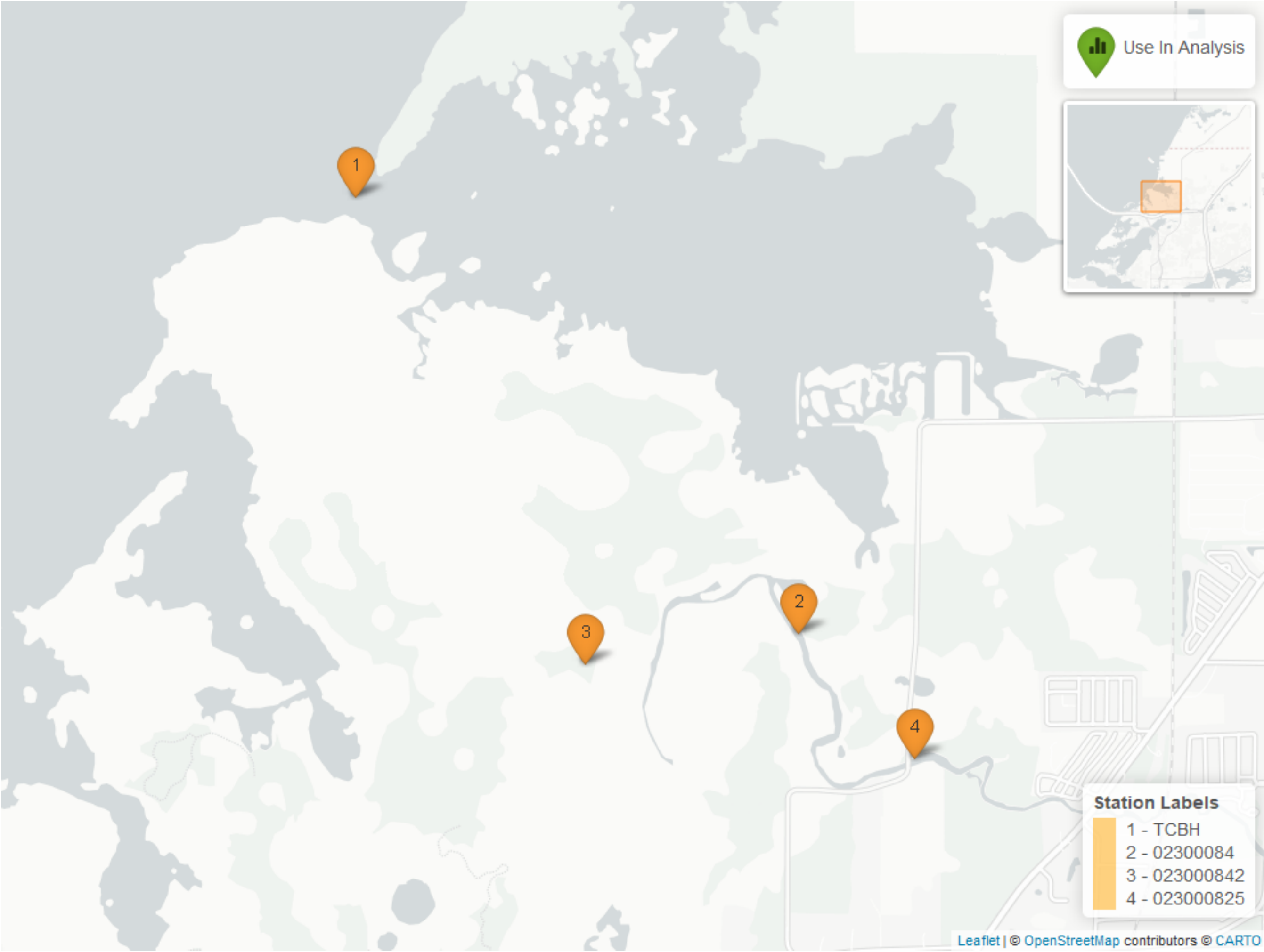

Figure 19: Map showing continuous water quality sampling locations within the boundaries of *Terra Ceia Aquatic Preserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

pH - Continuous

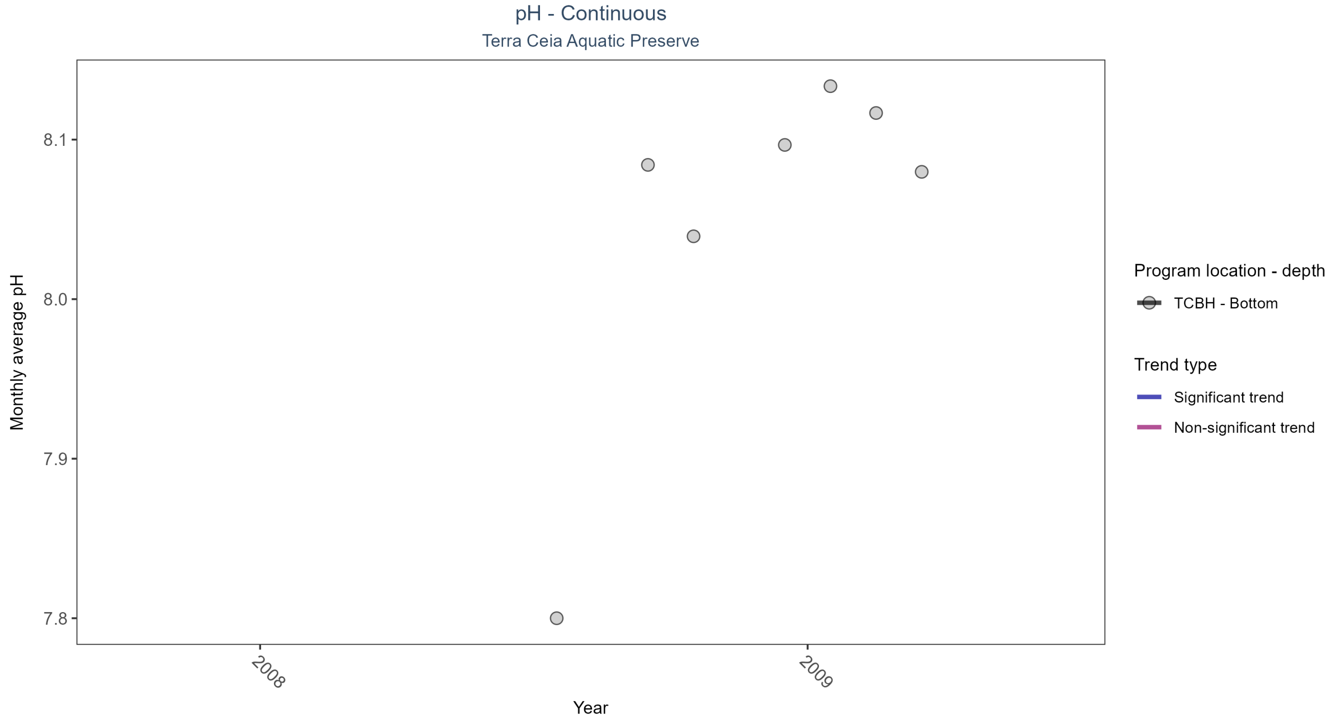

Figure 20: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 25: Seasonal Kendall-Tau Results for pH - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
TCBH	Insufficient data to calculate trend	8306	2	2008 - 2009	8.1	-	-	-	-

There was insufficient data to fit a model for one location.

Dissolved Oxygen Saturation - Continuous

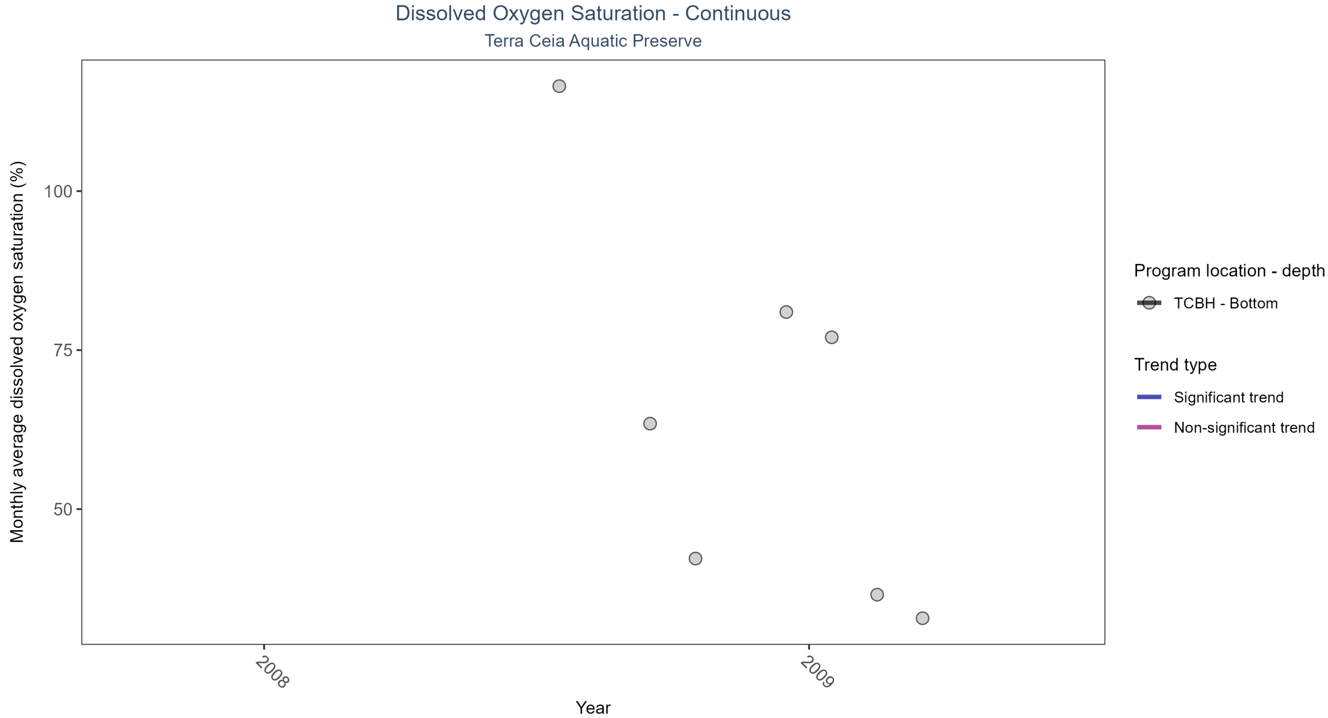

Figure 21: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 26: Seasonal Kendall-Tau Results for Dissolved Oxygen Saturation - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
TCBH	Insufficient data to calculate trend	8153	2	2008 - 2009	59.5	-	-	-	-

There was insufficient data to fit a model for one location.

Dissolved Oxygen - Continuous

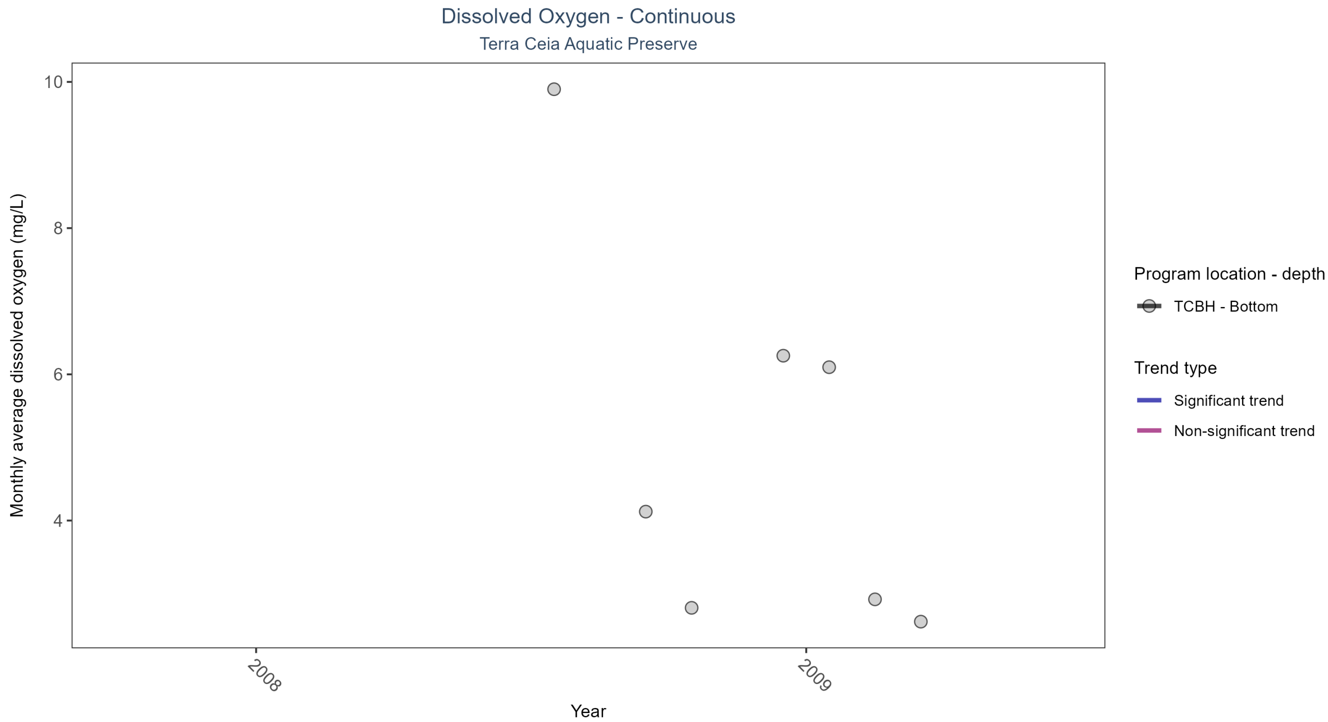

Figure 22: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 27: Seasonal Kendall-Tau Results for Dissolved Oxygen - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
TCBH	Insufficient data to calculate trend	8153	2	2008 - 2009	4.2	-	-	-	-

There was insufficient data to fit a model for one location.

Turbidity - Continuous

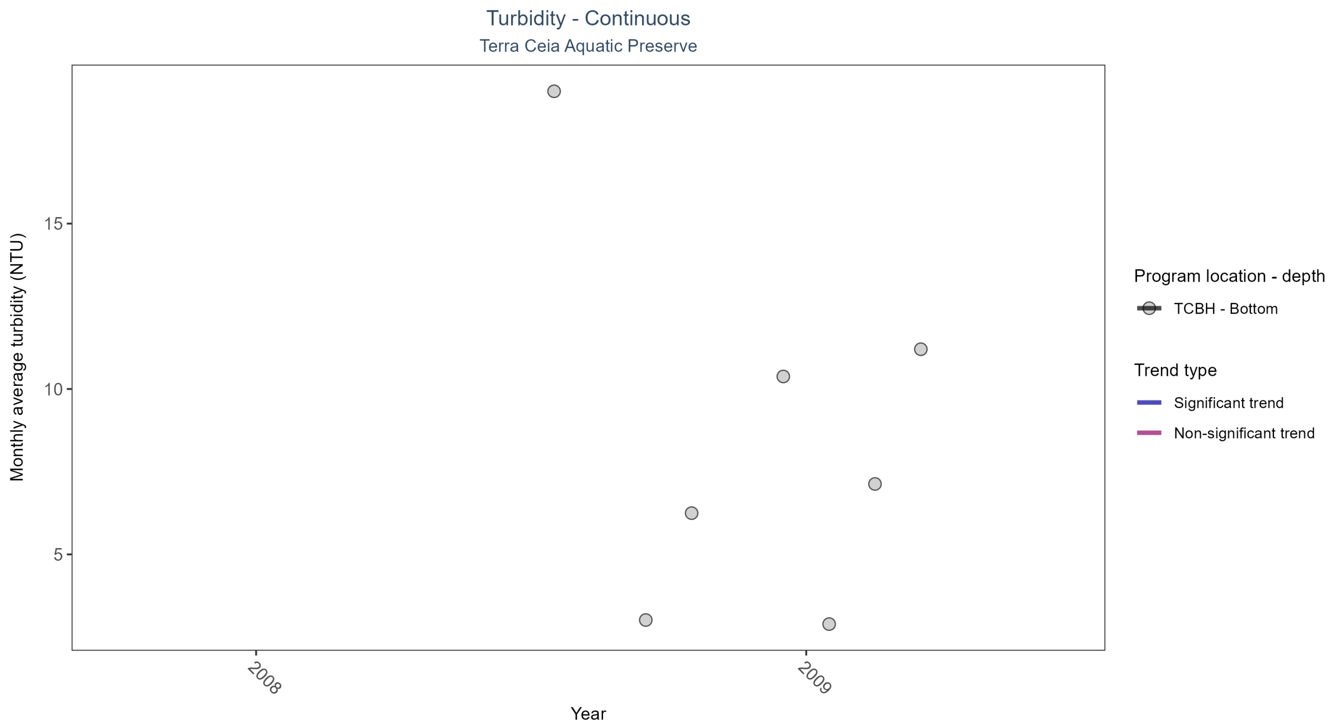

Figure 23: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 28: Seasonal Kendall-Tau Results for Turbidity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
TCBH	Insufficient data to calculate trend	8263	2	2008 - 2009	2	-	-	-	-

There was insufficient data to fit a model for one location.

Water Temperature - Continuous

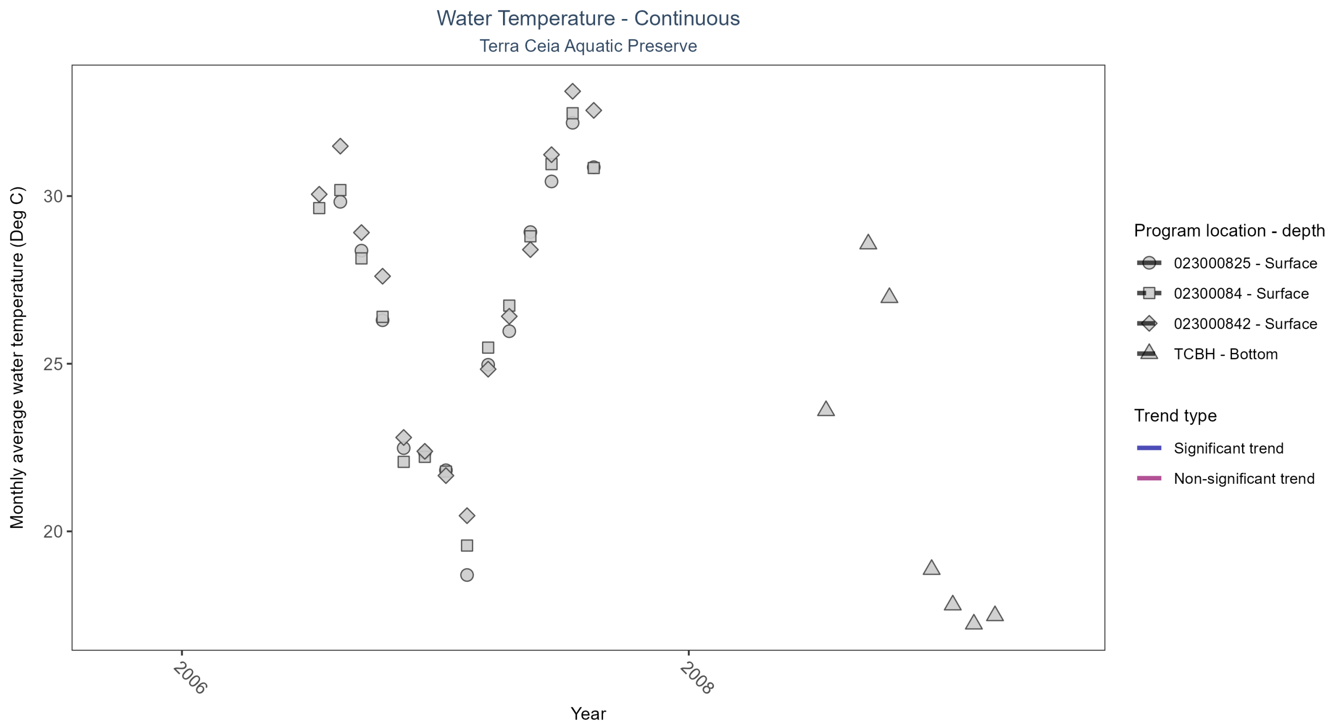

Figure 24: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for Water Temperature - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
023000825	Insufficient data to calculate trend	659	2	2006 - 2007	26.6	-	-	-	-
02300084	Insufficient data to calculate trend	717	2	2006 - 2007	27.5	-	-	-	-
023000842	Insufficient data to calculate trend	571	2	2006 - 2007	27.7	-	-	-	-
TCBH	Insufficient data to calculate trend	8305	2	2008 - 2009	20.0	-	-	-	-

There was insufficient data to fit a model for four locations.

Salinity - Continuous

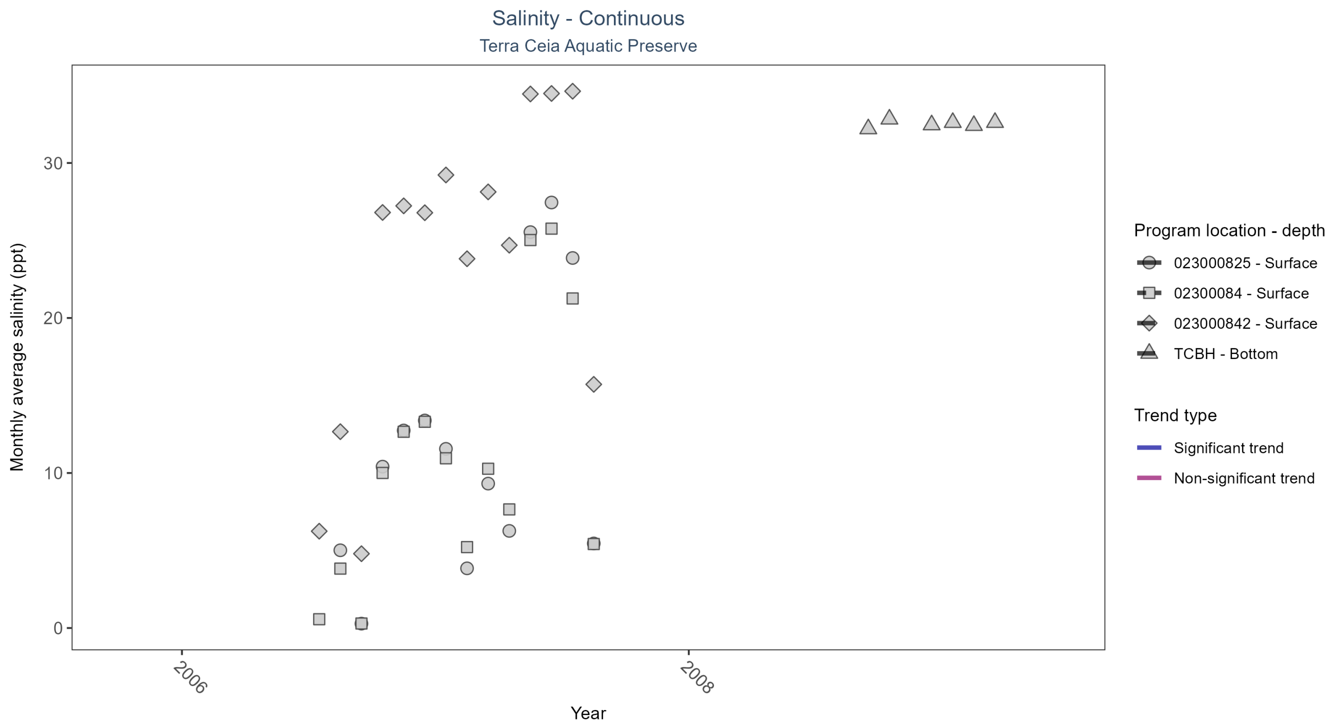

Figure 25: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 30: Seasonal Kendall-Tau Results for Salinity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
023000825	Insufficient data to calculate trend	645	2	2006 - 2007	12.0	-	-	-	-
02300084	Insufficient data to calculate trend	696	2	2006 - 2007	11.0	-	-	-	-
023000842	Insufficient data to calculate trend	578	2	2006 - 2007	28.0	-	-	-	-
TCBH	Insufficient data to calculate trend	8304	2	2008 - 2009	32.6	-	-	-	-

There was insufficient data to fit a model for four locations.

Specific Conductivity - Continuous

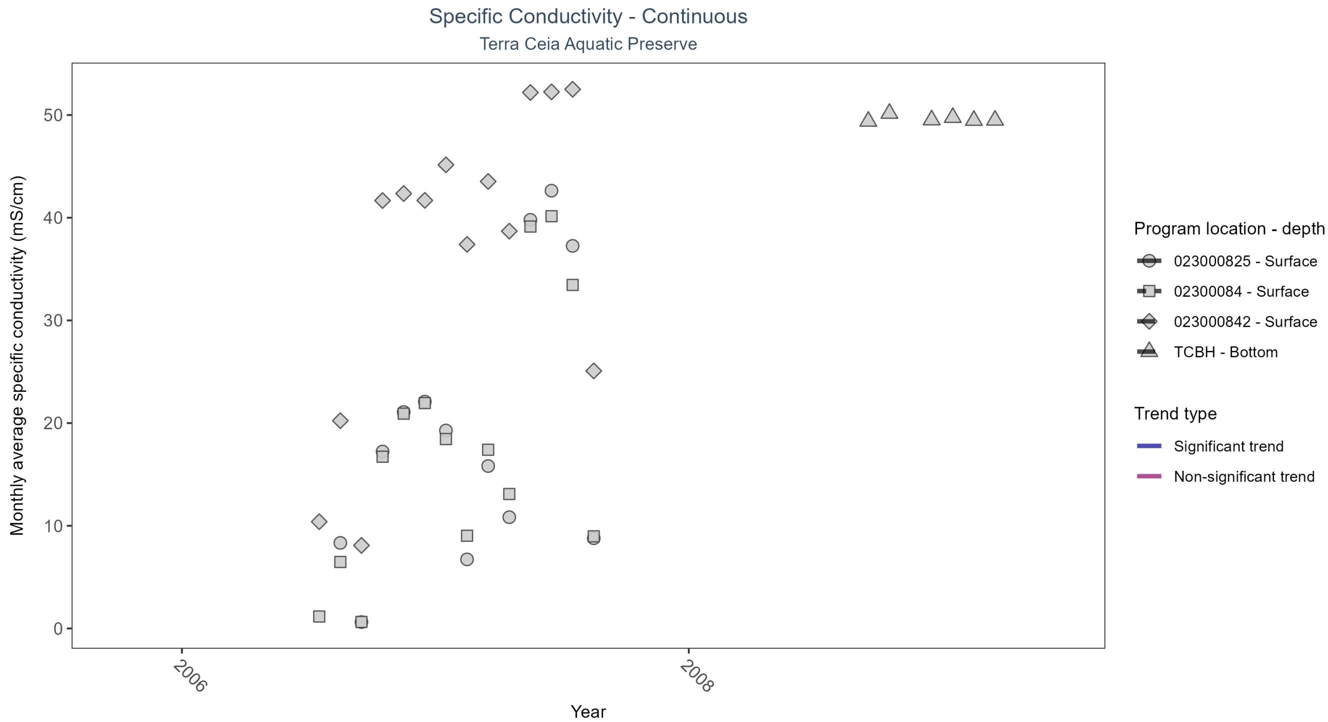

Figure 26: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 31: Seasonal Kendall-Tau Results for Specific Conductivity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
023000825	Insufficient data to calculate trend	710	2	2006 - 2007	18.60	-	-	-	-
02300084	Insufficient data to calculate trend	775	2	2006 - 2007	17.00	-	-	-	-
023000842	Insufficient data to calculate trend	669	2	2006 - 2007	43.10	-	-	-	-
TCBH	Insufficient data to calculate trend	8305	2	2008 - 2009	49.71	-	-	-	-

There was insufficient data to fit a model for four locations.

Submerged Aquatic Vegetation

The data file used is: **All__SAV__Parameters-2026-Jun-04.txt**

Submerged aquatic vegetation (*SAV*) refers to plants and plant-like macroalgae species that live entirely underwater. The two primary categories of *SAV* inhabiting Florida estuaries are *benthic macroalgae* and *seagrasses*. They often grow together in dense beds or meadows that carpet the seafloor. *Macroalgae* include multicellular species of green, red and brown algae that often live attached to the substrate by a holdfast. They tend to grow quickly and can tolerate relatively high nutrient levels, making them a threat to seagrasses and other benthic habitats in areas with poor water quality. In contrast, *seagrasses* are grass-like, vascular, flowering plants that are attached to the seafloor by extensive root systems. *Seagrasses* occur throughout the coastal areas of Florida, including protected bays and lagoons as well as deeper offshore waters on the continental shelf. *Seagrasses* have taken advantage of the broad, shallow shelf and clear water to produce two of the most extensive seagrass beds anywhere in continental North America.

Parameters

Percent Cover measures the fraction of an area of seafloor that is covered by SAV, usually estimated by evaluating multiple small areas of seafloor. Percent cover is often estimated for total SAV, individual types of vegetation (seagrass, attached algae, drift algae) and individual species.

Frequency of Occurrence was calculated as the number of times a taxon was observed in a year divided by the number of sampling events, multiplied by 100. Analysis is conducted at the quadrat level and is inclusive of all quadrats (i.e., quadrats evaluated using Braun-Blanquet, modified Braun-Blanquet, and percent cover."

Species

Turtle grass (*Thalassia testudinum*) is the largest of the Florida seagrasses, with longer, thicker blades and deeper root structures than any of the other seagrasses. It is considered a climax seagrass species.

Shoal grass (*Halodule wrightii*) is an early colonizer of vegetated areas and usually grows in water too shallow for other species except *widgeon grass*. It can often tolerate larger salinity ranges than other seagrass species. *Shoal grass* is characterized by thin, flat blades, that are narrower than *turtle grass* blades.

Manatee grass (*Syringodium filiforme*) is easily recognizable because its leaves are thin and cylindrical instead of the flat, ribbon-like form shared by many other seagrass species. The leaves can grow up to half a meter in length. *Manatee grass* is usually found in mixed seagrass beds or small, dense monospecific patches.

Widgeon grass (*Ruppia maritima*) grows in both fresh and salt water and is widely distributed throughout Florida's estuaries in less saline areas, particularly in inlets along the east coast. This species resembles *shoal grass* in certain environments but can be identified by the pointed tips of its leaves.

Three species of *Halophila spp.* are found in Florida - **Star grass** (*Halophila engelmannii*), **Paddle grass** (*Halophila decipiens*), and **Johnson's seagrass** (*Halophila johnsonii*). These are smaller, more fragile seagrasses than other Florida species and are considered ephemeral. They grow along a single long rhizome, with short blades. These species are not well-studied, although surveys are underway to define their ecological roles.

Notes

Star grass, *Paddle grass*, and *Johnson's seagrass* will be grouped together and listed as **Halophila spp.** in the following managed areas. This is because several surveys did not specify to the species level:

- Banana River Aquatic Preserve
- Indian River-Malabar to Vero Beach Aquatic Preserve
- Indian River-Vero Beach to Ft. Pierce Aquatic Preserve
- Jensen Beach to Jupiter Inlet Aquatic Preserve
- Loxahatchee River-Lake Worth Creek Aquatic Preserve
- Mosquito Lagoon Aquatic Preserve

- Biscayne Bay Aquatic Preserve
- Florida Keys National Marine Sanctuary

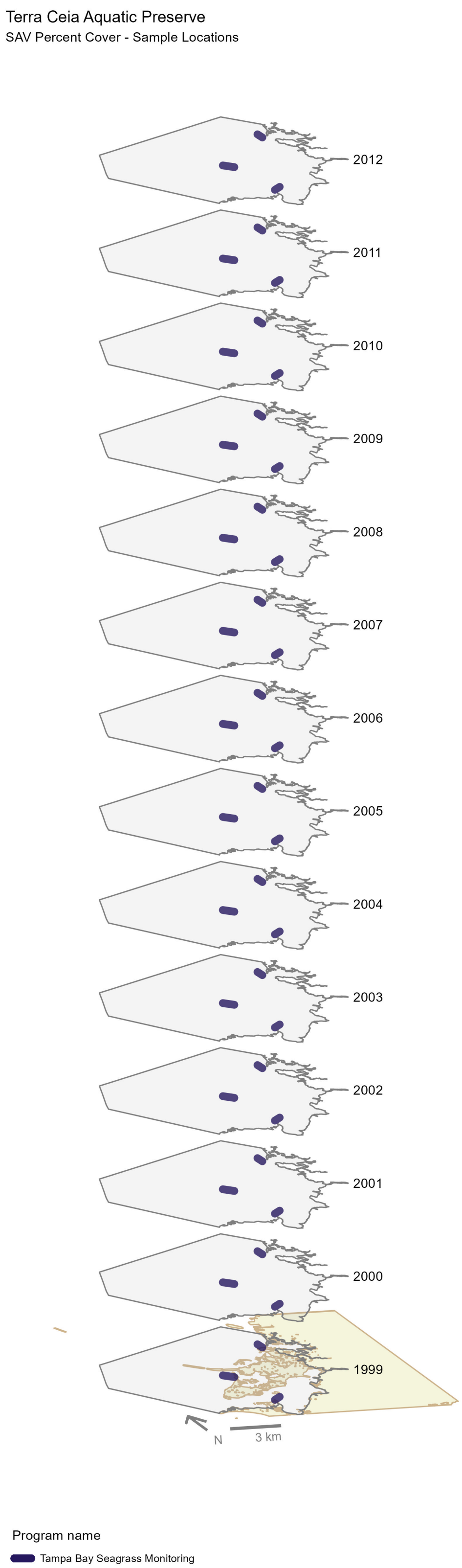

Figure 27: Maps showing the temporal scope of SAV sampling sites within the boundaries of *Terra Ceia Aquatic Preserve* by program name.

Click here to view spatio-temporal plots on GitHub.

Sampling locations by Program:

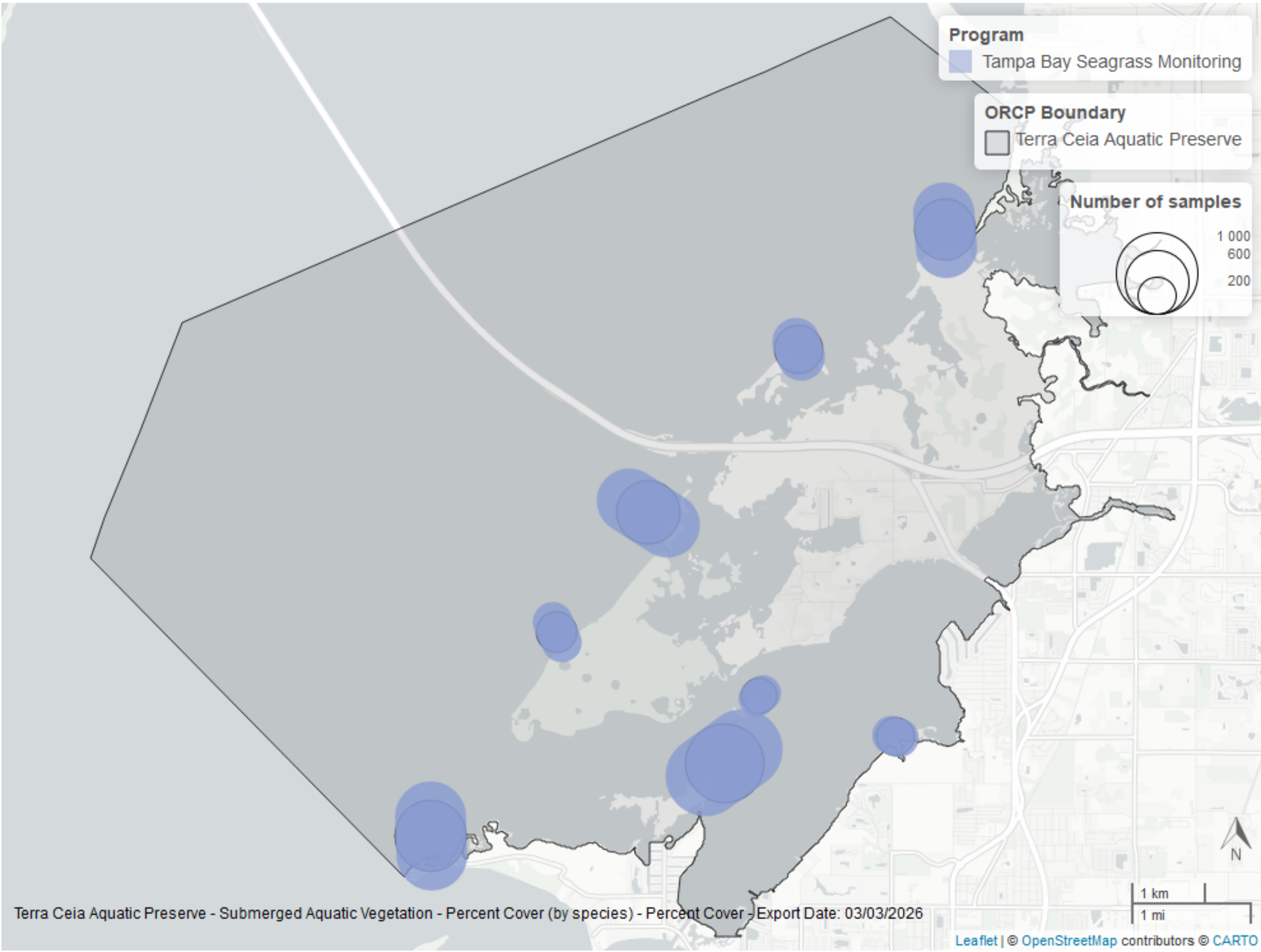

Figure 28: Map showing SAV sampling sites within the boundaries of *Terra Ceia Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Table 32: Program Information for Submerged Aquatic Vegetation

ProgramID	*N-Data*	*YearMin*	*YearMax*	*method*	*Sample Locations*
565	4017	1999	2024	Braun Blanquet	8

Program names:

565 - Tampa Bay Seagrass Monitoring[9]

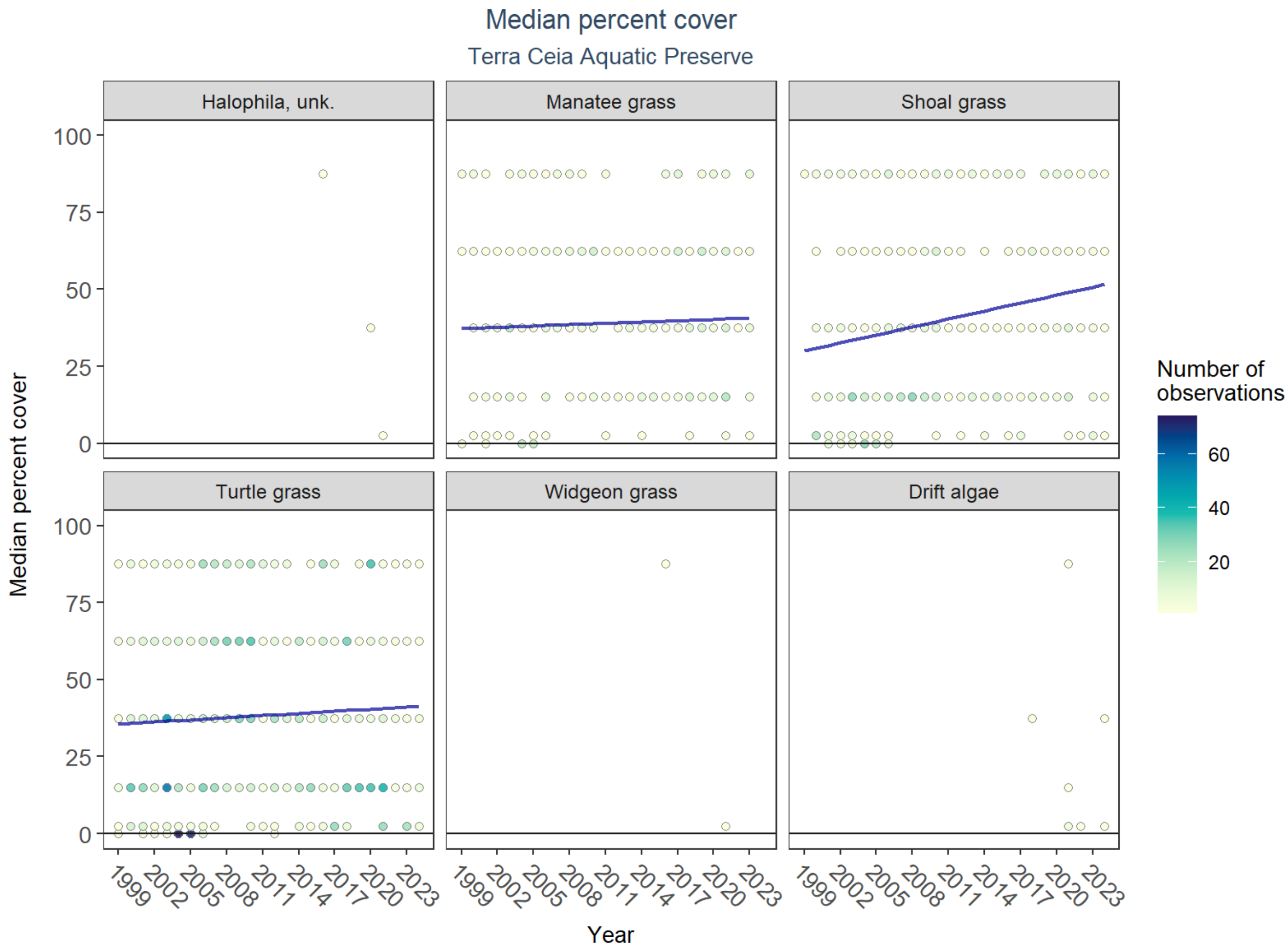

Figure 29: Scatter plots of median percent cover of submerged aquatic vegetation over time by group. Plots for time series that included five or more years of observations show the estimated trend as a blue line.

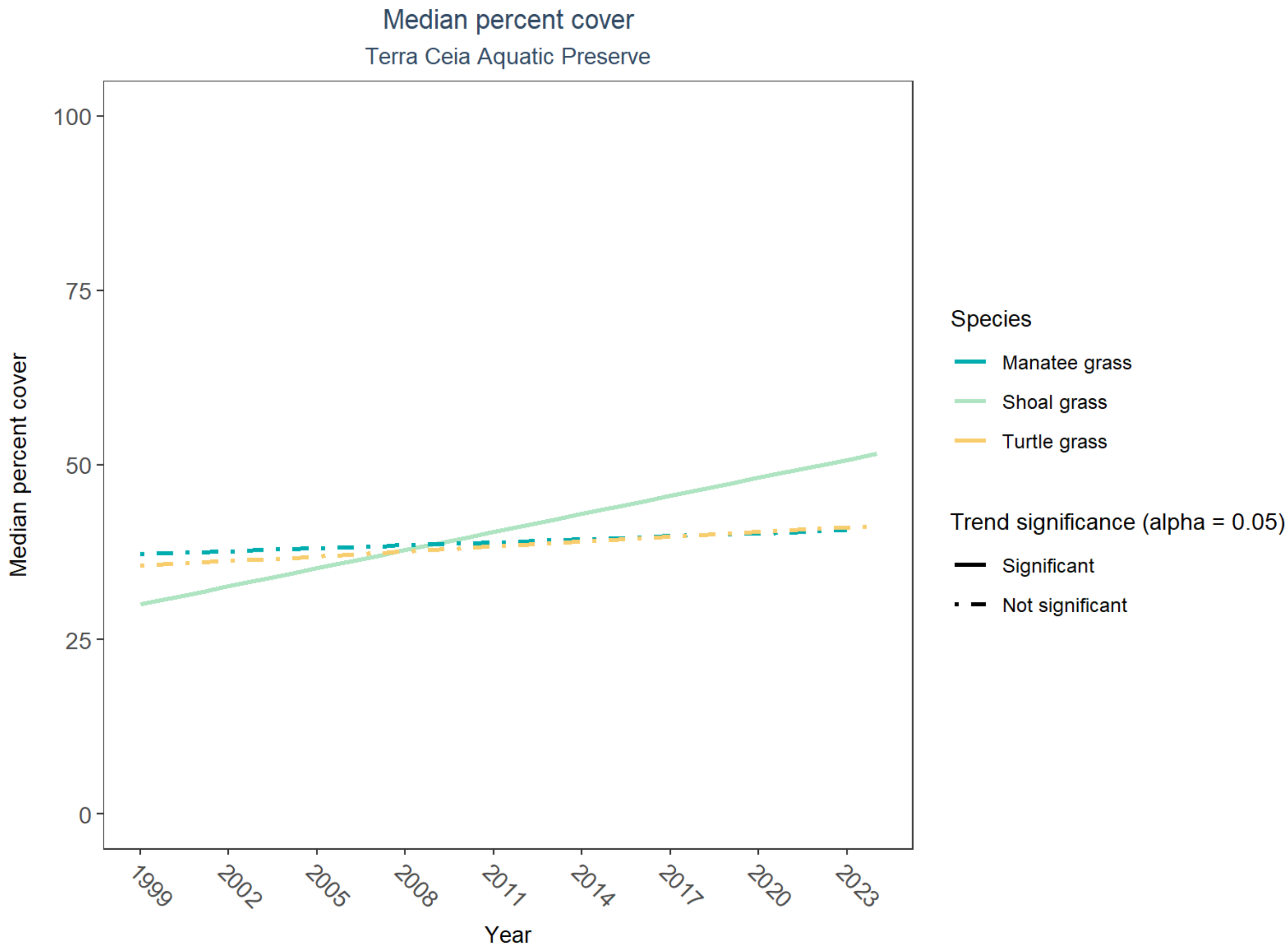

Figure 30: Trend lines of median percent cover of submerged aquatic vegetation over time for species that had five or more years of observations. Line type represents significance (solid) or non-significance (dashed) of the trends.

Table 33: Percent Cover Trend Analysis for Terra Ceia Aquatic Preserve

Species	*Statistical Trend*	*Period of Record*	*LME Intercept*	*LME Slope*	*P*
Drift algae	Insufficient data	-	-	-	-
Shoal grass	Increasing trend	1999 - 2024	25.73515	0.862849	0.0041757
Widgeon grass	Insufficient data	-	-	-	-
Manatee grass	No detectable trend	1999 - 2023	36.53748	0.141925	0.7081145
Turtle grass	No detectable trend	1999 - 2024	34.43132	0.229663	0.4469083
Halophila, unk.	Insufficient data	-	-	-	-

An annual increase in percent cover was observed for shoal grass (0.86%). No detectable change in percent cover was observed for manatee grass and turtle grass. Trends in percent cover could not be evaluated for unknown Halophila, widgeon grass, and drift algae due to insufficient data.

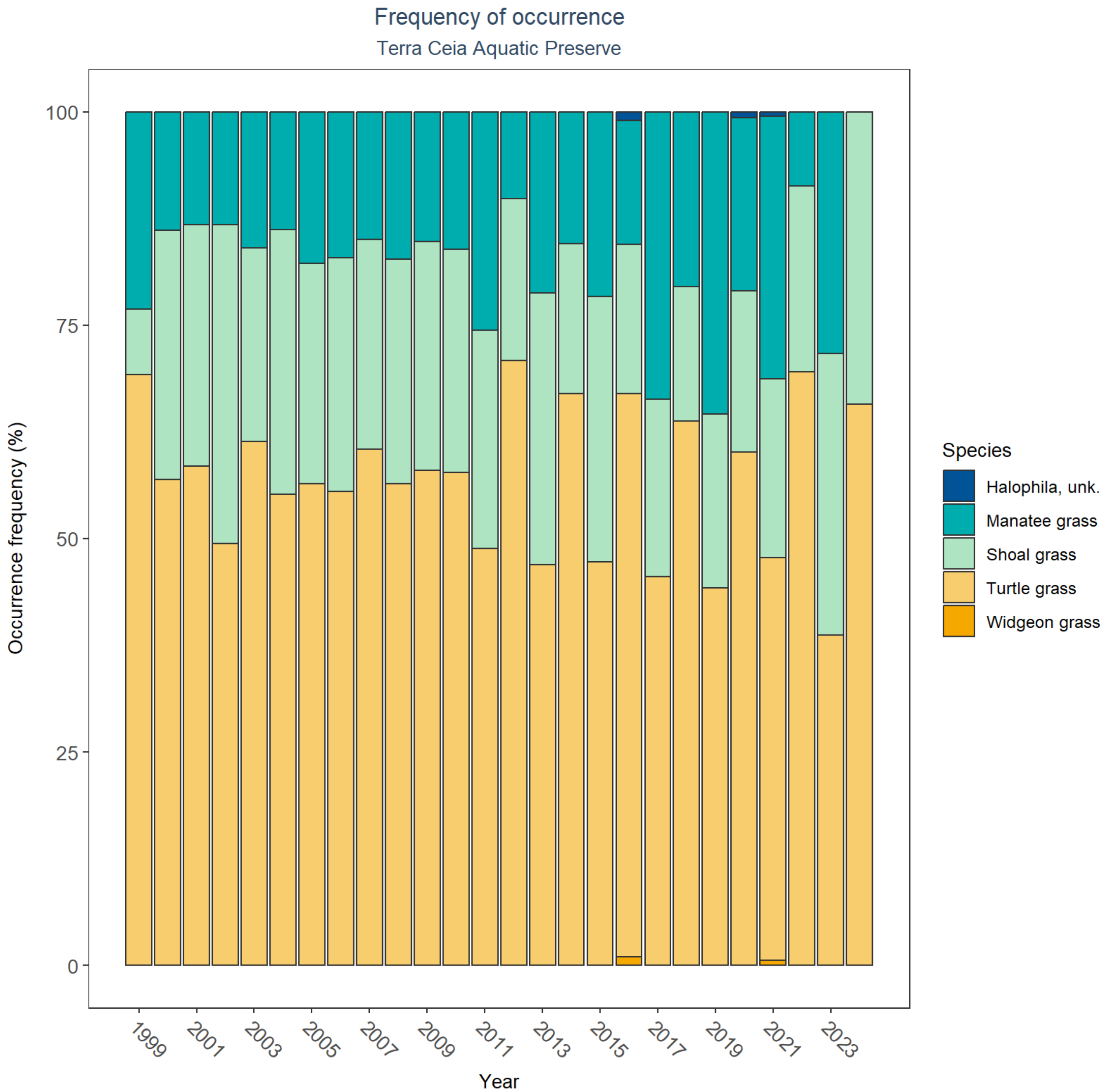

Figure 31: Bar plot of submerged aquatic vegetation species occurrence frequency over time by group.

SAV Water Column Analysis

The following parameters are available for Terra Ceia Aquatic Preserve within the SAV_WC_Report:

- Chlorophyll a
- Dissolved Oxygen
- Dissolved Oxygen Saturation
- pH
- Salinity
- Secchi Depth

- Water Temperature
- Total Nitrogen
- Total Suspended Solids
- Turbidity

Access the reports here: DRAFT_SAV_WC_Report.pdf

Nekton

The data file used is: **All_NEKTON_Parameters-2026-Jun-04.txt**

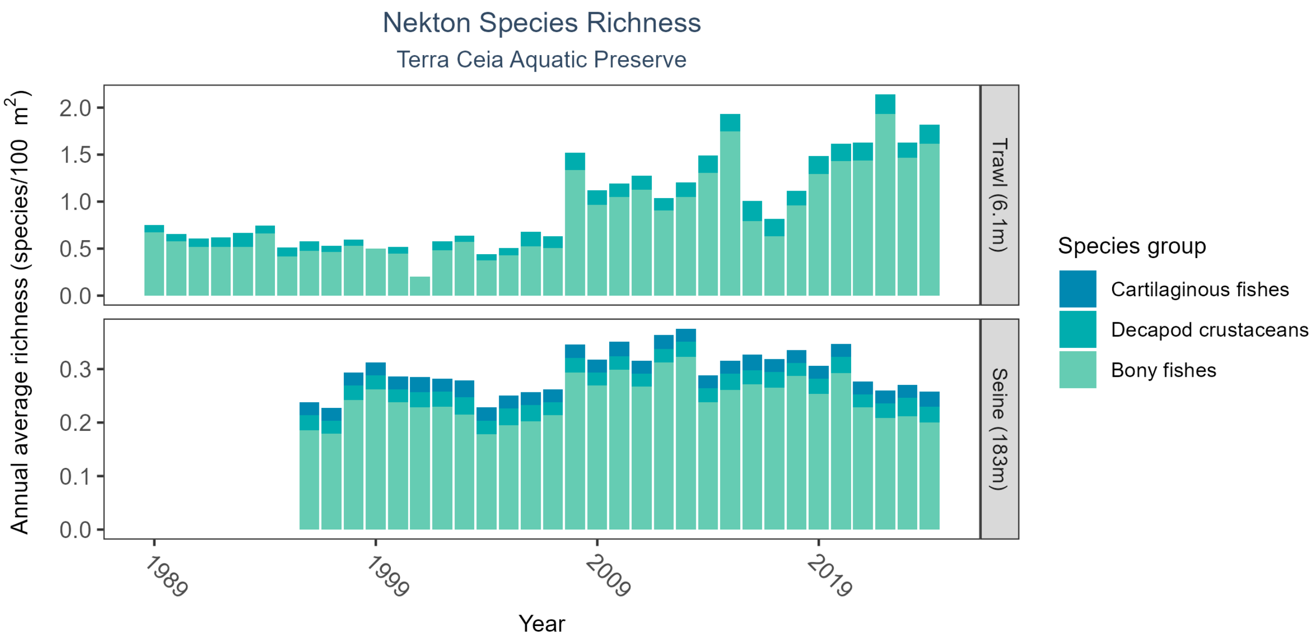

Figure 32: Bar graph(s) of annual average nekton richness over time for species groups occurring in at least 1% of samples. The bar colors represent species groups including bony fishes, cartilaginous fishes, decapod crustaceans (e.g., shrimps, crabs, and lobsters), and cephalopods (e.g., squid). Gear types and sizes are indicated in the panel label.

Table 34: Nekton Species Richness

Gear Type	*No. of Samples*	*No. of Years with Data*	*Period of Record*	*Median No. of Taxa*	*Mean No. of Taxa*
Trawl (6.1 m)	1382	36	1989 - 2024	0.32	0.65
Seine (183 m)	1543	29	1996 - 2024	0.17	0.17

The median annual number of taxa was 0.17 based on 1,543 observations collected by 183-meter seine between 1996 and 2024, and the median annual number of taxa was 0.32 based on 1,382 observations collected by 6.1-meter trawl between 1989 and 2024.

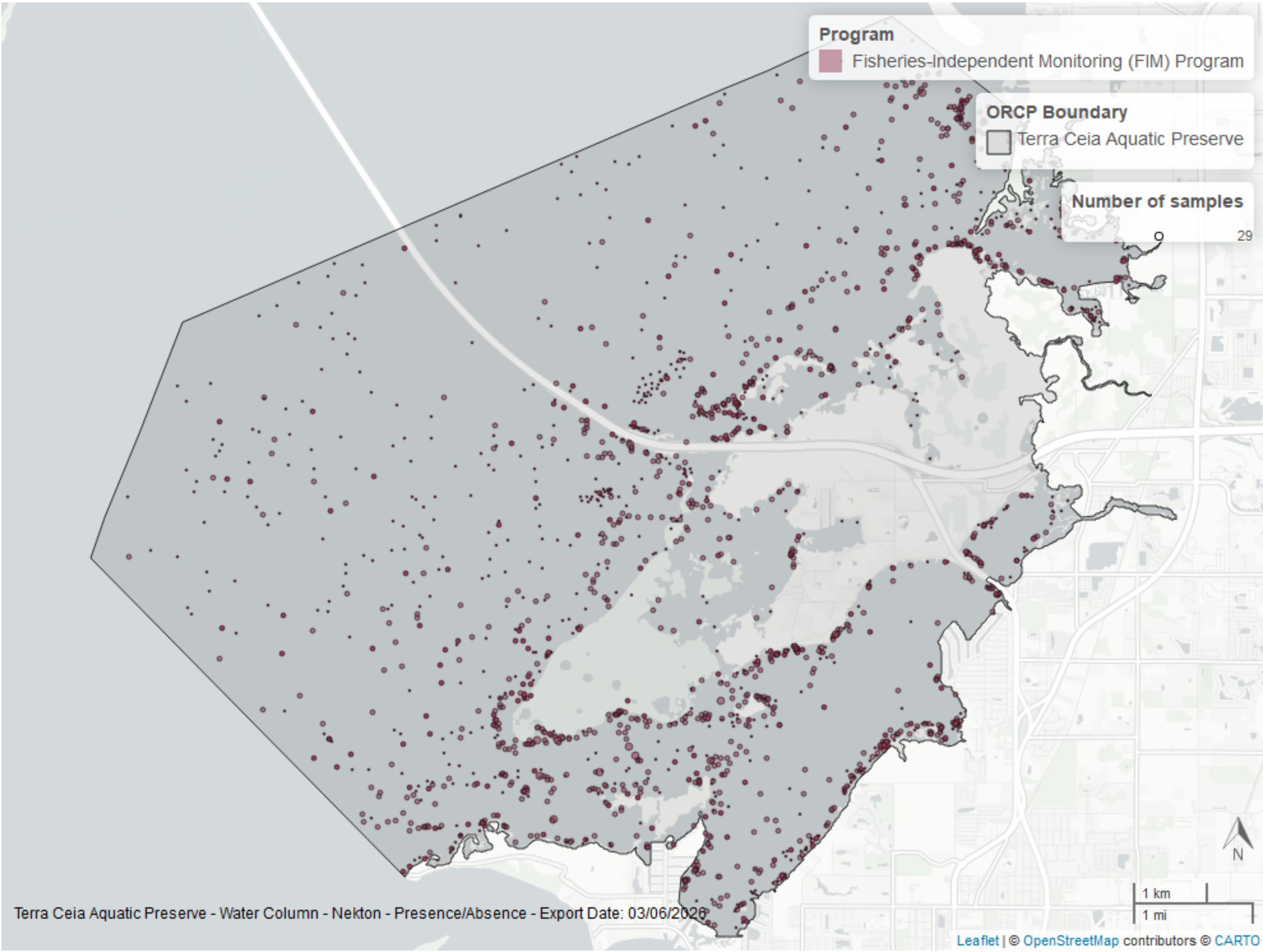

Figure 33: Map showing location of nekton sampling locations within the boundaries of *Terra Ceia Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species list

Acanthophora sp.[1]	Eucinostomus argenteus[2]	No grass in quadrat[1]
Acanthostracion quadricornis[2]	Eucinostomus gula[2]	Ocyurus chrysurus[2]
Achirus lineatus[2]	Eucinostomus harengulus[2]	Ogcocephalus cubifrons[2]
Aetobatus narinari[2]	Eucinostomus spp.[2]	Oligoplites saurus[2]
Albula vulpes[2]	Eugerres plumieri[2]	Ophidiidae spp.[2]
Alpheidae spp.[2]	Floridichthys carpio[2]	Opisthonema oglinum[2]
Aluterus schoepfii[2]	Fundulus grandis[2]	Opistognathus spp.[2]
Aluterus scriptus[2]	Fundulus similis[2]	Opsanus beta[2]
Anarchopterus criniger[2]	Gerres cinereus[2]	Oreochromis aureus[2]
Anchoa cubana[2]	Gobiosoma bosc[2]	Oreochromis niloticus[2]
Anchoa hepsetus[2]	Gobiosoma longipala[2]	Orthopristis chrysoptera[2]
Anchoa mitchilli[2]	Gobiosoma robustum[2]	Ostraciidae spp.[2]
Anchoa spp.[2]	Gobiosoma spp.[2]	Other green algae[1]
Ancylopsetta quadrocellata[2]	Gracilaria sp.[1]	Paraclinus marmoratus[2]
Archosargus probatocephalus[2]	Haemulon aurolineatum[2]	Paralichthys albigutta[2]
Archosargus rhomboidalis[2]	Haemulon plumierii[2]	Penaeus duorarum[2]
Argopecten irradians	Halichoeres bivittatus[2]	Poecilia latipinna[2]
Argopecten spp.	Halodule wrightii[1]	Pogonias cromis[2]
Ariopsis felis[2]	Halophila sp.[1]	Pomatomus saltatrix[2]
Astroscopus ygraecum[2]	Harengula jaguana[2]	Portunidae spp.[2]
Bagre marinus[2]	Hemiramphidae spp.[2]	Portunus spp.[2]
Bairdiella chrysoura[2]	Hemiramphus brasiliensis[2]	Prionotus scitulus[2]
Brevoortia spp.[2]	Hippocampus erectus[2]	Prionotus tribulus[2]
Calamus arctifrons[2]	Hippocampus zosterae[2]	Pseudocrenilabrinae[2]
Calamus bajonado[2]	Hypleurochilus caudovittatus[2]	Rachycentron canadum[2]
Calamus penna[2]	Hypleurochilus geminatus[2]	Rhinoptera bonasus[2]
Calamus spp.[2]	Hypnea[1]	Rimapenaeus constrictus[2]
Callinectes ornatus[2]	Hyporhamphus meeki[2]	Rimapenaeus spp.[2]
Callinectes sapidus[2]	Hyporhamphus spp.[2]	Ruppia maritima[1]
Callinectes spp.[2]	Hyporhamphus unifasciatus[2]	Sardinella aurita[2]
Caranx crysos[2]	Hypsoblennius hentz[2]	Sarotherodon melanotheron[2]
Caranx hippos[2]	Lachnolaimus maximus[2]	Sciaenops ocellatus[2]
Caranx latus[2]	Lactophrys trigonus[2]	Scomberomorus maculatus[2]
Caranx spp.[2]	Lagocephalus laevigatus[2]	Scorpaena brasiliensis[2]
Carcharhinus limbatus[2]	Lagodon rhomboides[2]	Selene vomer[2]
Caulerpa mexicana[1]	Leiostomus xanthurus[2]	Serraniculus pumilio[2]
Caulerpa prolifera[1]	Lepisosteus osseus[2]	Serranus subligarius[2]
Caulerpa sertularioides[1]	Limulus polyphemus	Sicyonia brevirostris[2]
Centropomus undecimalis[2]	Lucania parva[2]	Sicyonia laevigata[2]
Centropristis striata[2]	Lutjanus analis[2]	Sicyonia typica[2]
Chaetodipterus faber[2]	Lutjanus griseus[2]	Sphoeroides nephelus[2]
Chasmodes saburrae[2]	Lutjanus synagris[2]	Sphoeroides spengleri[2]
Chelonia mydas[2]	Lyngbya sp.	Sphyraena barracuda[2]
Chilomycterus schoepfii[2]	Malaclemys terrapin[2]	Sphyraena borealis[2]
Chloroscombrus chrysurus[2]	Menidia spp.[2]	Sphyrna tiburo[2]
Citharichthys macrops[2]	Menippe mercenaria[2]	Stomolophus meleagris
Cynoscion arenarius[2]	Menippe spp.[2]	Strongylura marina[2]
Cynoscion nebulosus[2]	Menticirrhus americanus[2]	Strongylura notata[2]
Cyprinodon variegatus[2]	Menticirrhus saxatilis[2]	Strongylura timucu[2]
Decapterus punctatus[2]	Menticirrhus spp.[2]	Syacium papillosum[2]
Diapterus auratus[2]	Microgobius gulosus[2]	Symphurus plagiusa[2]
Diodon spp.[2]	Microgobius thalassinus[2]	Syngnathus floridae[2]
Diplectrum formosum[2]	Micropogonias undulatus[2]	Syngnathus louisianae[2]
Diplectrum spp.[2]	Monacanthus ciliatus[2]	Syngnathus scovelli[2]

Diplodus holbrookii[2]	Monacanthus spp.[2]	Syngnathus spp.[2]
Dorosoma petenense[2]	Mugil cephalus[2]	Synodus foetens[2]
Drift algae[1]	Mugil curema[2]	Syringodium filiforme[1]
Drift red algae[1]	Mugil spp.[2]	Thalassia testudinum[1]
Echeneis naucrates[2]	Mugil trichodon[2]	Trachinotus carolinus[2]
Echeneis neucratoides[2]	Mycteroperca microlepis[2]	Trachinotus falcatus[2]
Elops saurus[2]	Myrophis punctatus[2]	Trinectes maculatus[2]
Epinephelus itajara[2]	Negaprion brevirostris[2]	Tylosurus crocodilus[2]
Epinephelus morio[2]	Nicholsina usta[2]	Urophycis floridana[2]
Etropus crossotus[2]	No fish	Acanthophora sp.[1]

1 - Submerged Aquatic Vegetation, 2 - Nekton

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